



Physiology of Blood

RATNA KUSUMAWATI

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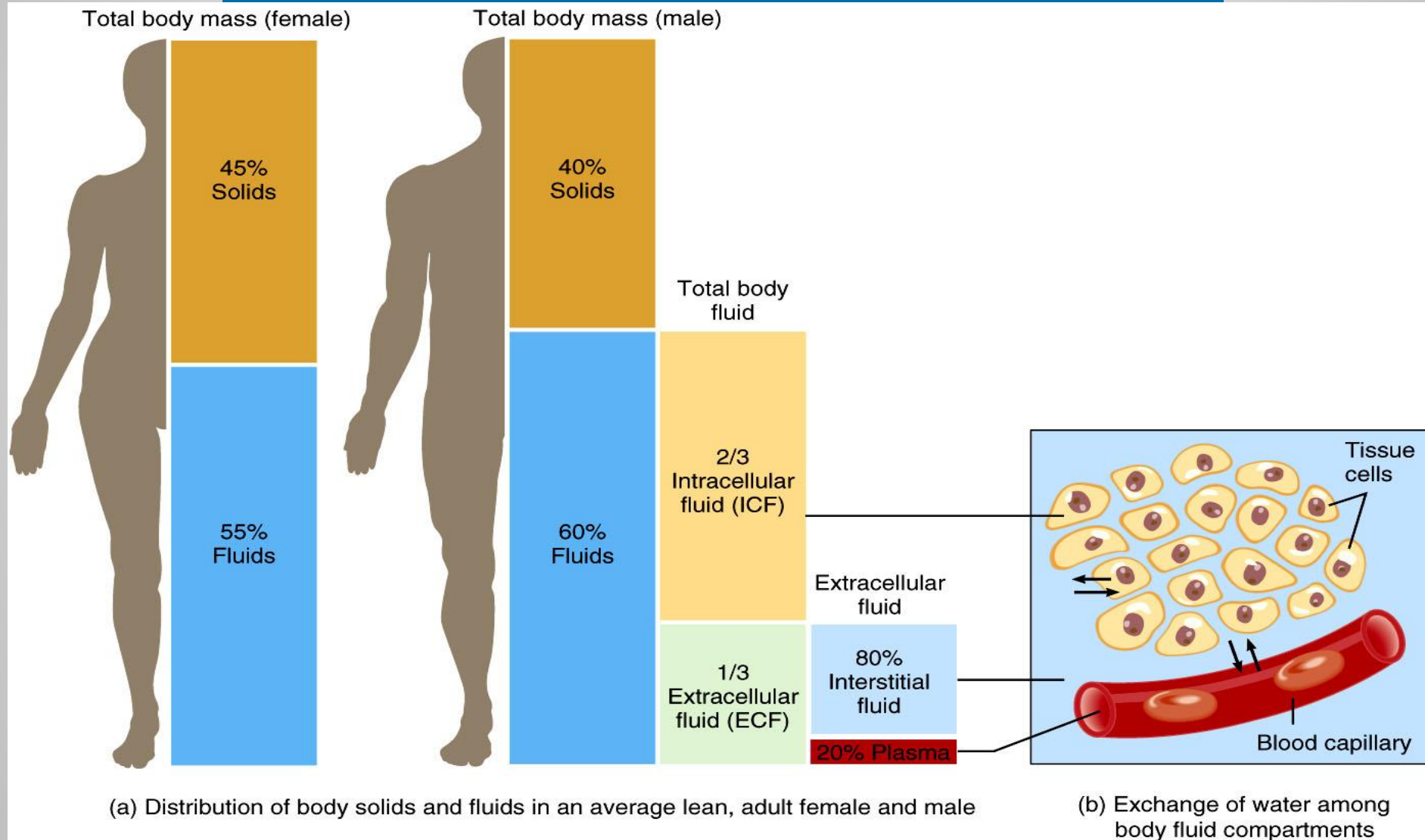
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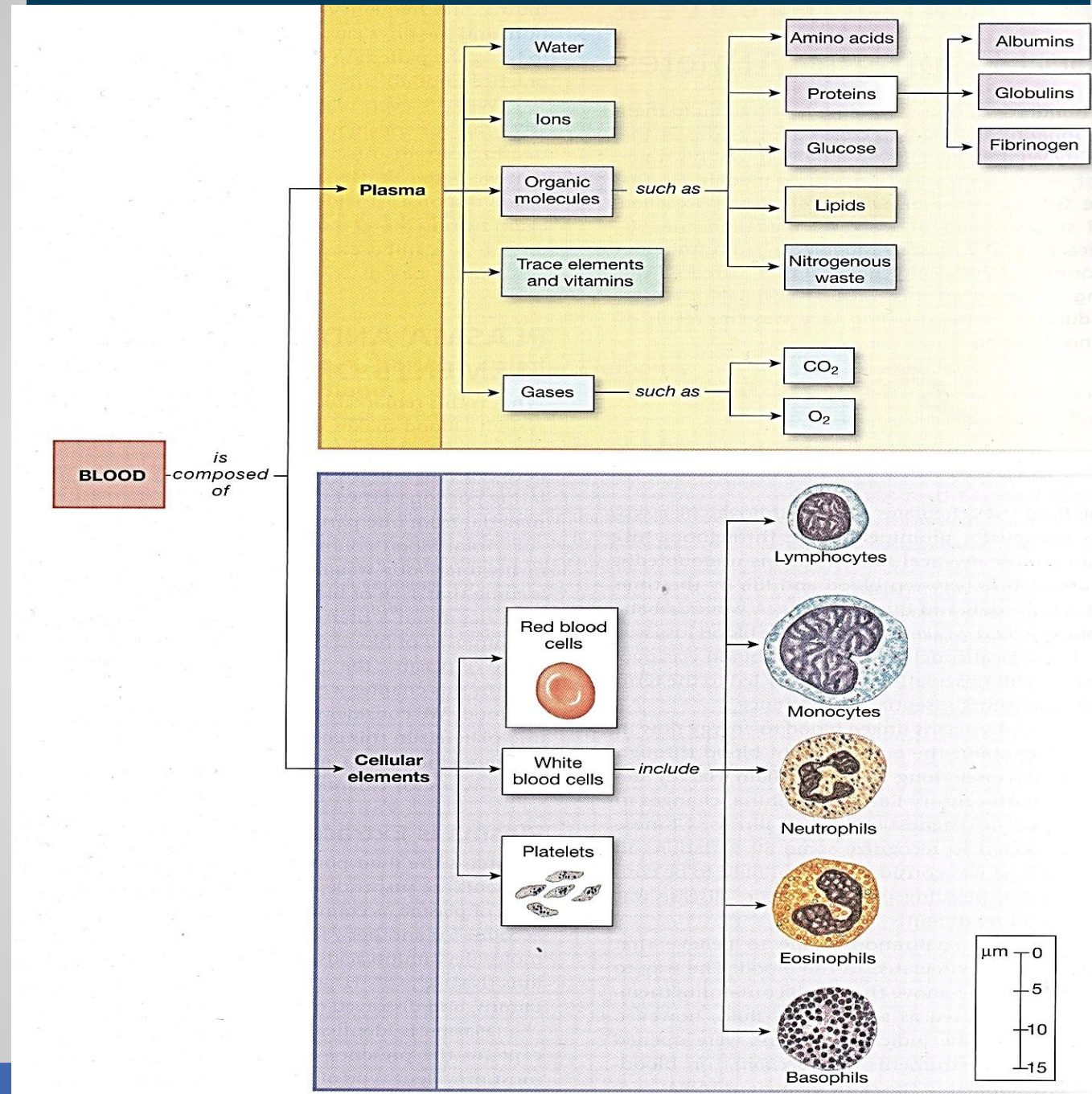


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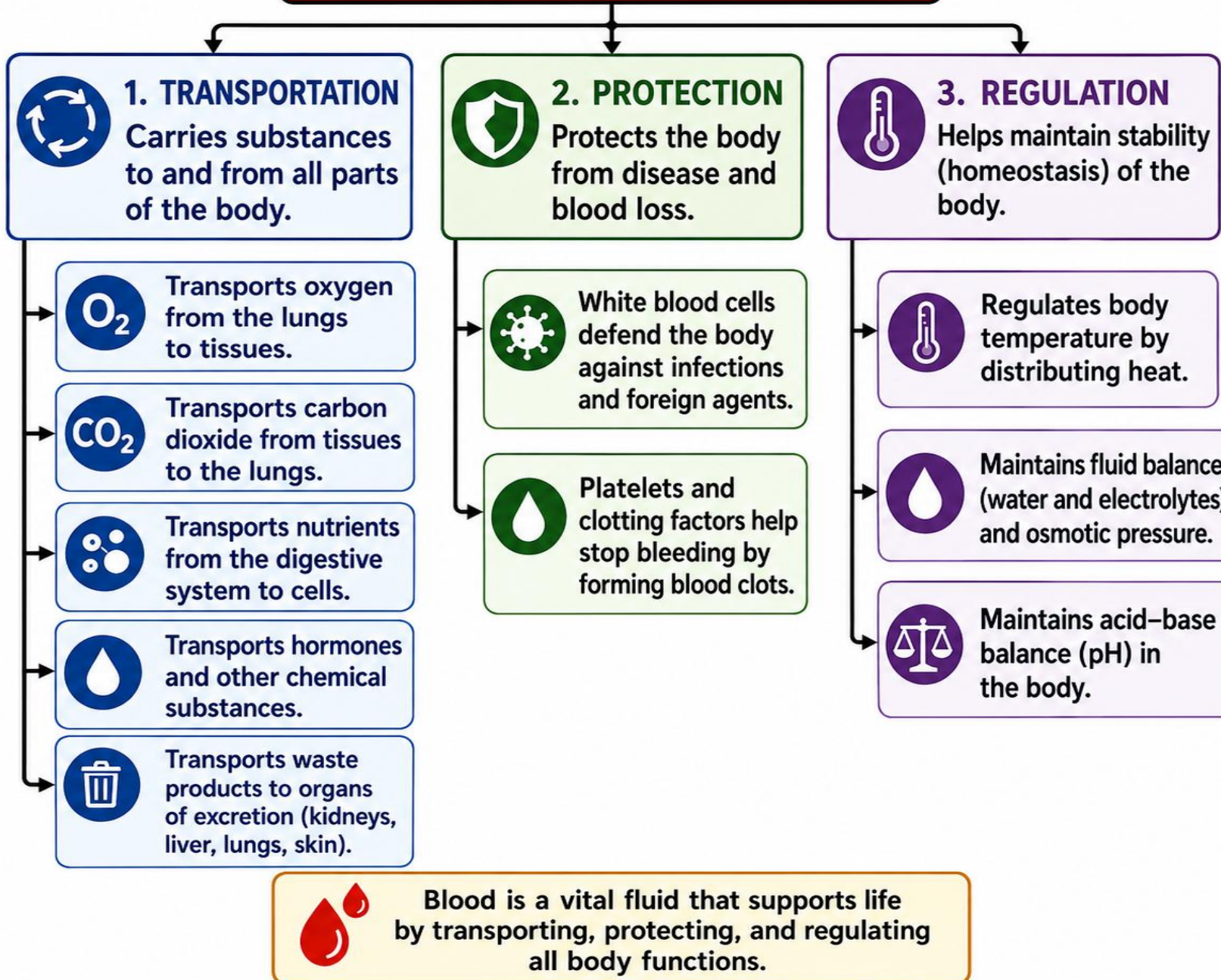
Body Fluid Compartments





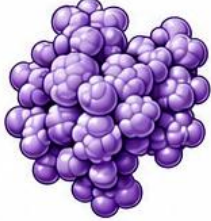
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FUNCTIONS OF BLOOD



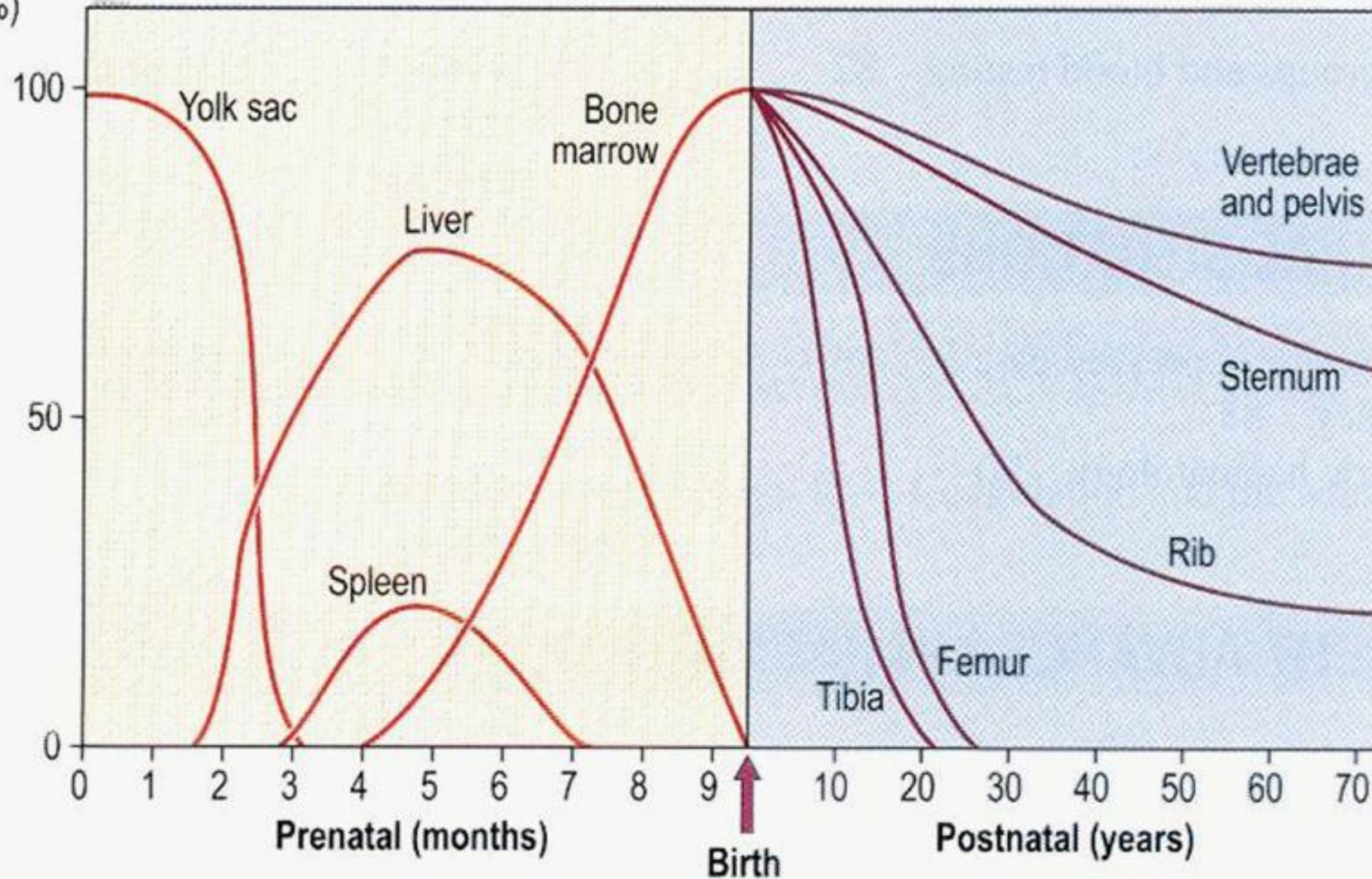
PROTEIN COMPONENTS IN BLOOD AND THEIR FUNCTIONS

Plasma (blood) proteins are a diverse group of proteins that play essential roles in maintaining homeostasis, protecting the body, and supporting many vital processes.

PROTEIN COMPONENT	MAIN COMPONENTS	FUNCTIONS
1. ALBUMIN (~ 55–60% of plasma proteins) 	• Most abundant plasma protein • Synthesized in the liver • Molecular weight: ~ 66 kDa	• Maintains oncotic (colloid osmotic) pressure and fluid balance • Transports various substances (fatty acids, bilirubin, hormones, drugs, vitamins) • Acts as an antioxidant • Serves as a reservoir of amino acids
2. GLOBULINS (~ 35–38% of plasma proteins) 	• Group of diverse proteins with different functions • Synthesized in the liver and lymphoid tissues • Include α, β, γ globulins	• Transport proteins (iron, lipids, hormones, vitamins) • Immune defense (antibodies) • Enzyme carriers and complement system components • Maintain oncotic pressure (less than albumin)
3. IMMUNOGLOBULINS (ANTIBODIES) (~ 15–20% of globulins) 	• Produced by B lymphocytes (plasma cells) • Classes: IgG, IgA, IgM, IgD, IgE	• Recognize and neutralize pathogens (bacteria, viruses, toxins) • Activate complement system • Provide humoral (antibody-mediated) immunity
4. FIBRINOGEN (~ 2–4% of plasma proteins) 	• Synthesized in the liver • Clotting factor I	• Converted by thrombin to fibrin • Forms fibrin mesh to stabilize blood clot • Essential for hemostasis
5. OTHER PLASMA PROTEINS (~ 1–2% of plasma proteins) Includes: transferrin, ceruloplasmin, complement proteins, clotting factors (except fibrinogen), enzymes, acute-phase proteins (e.g., CRP) 	• Synthesized in liver or other tissues • Include transport proteins, enzymes, complement proteins, and acute-phase proteins	• Transport metals (iron, copper) and other molecules • Participate in immune defense (complement system) • Involve in coagulation cascade • Increase during inflammation or infection

HEMATOPOESIS

Blood
production
(%)



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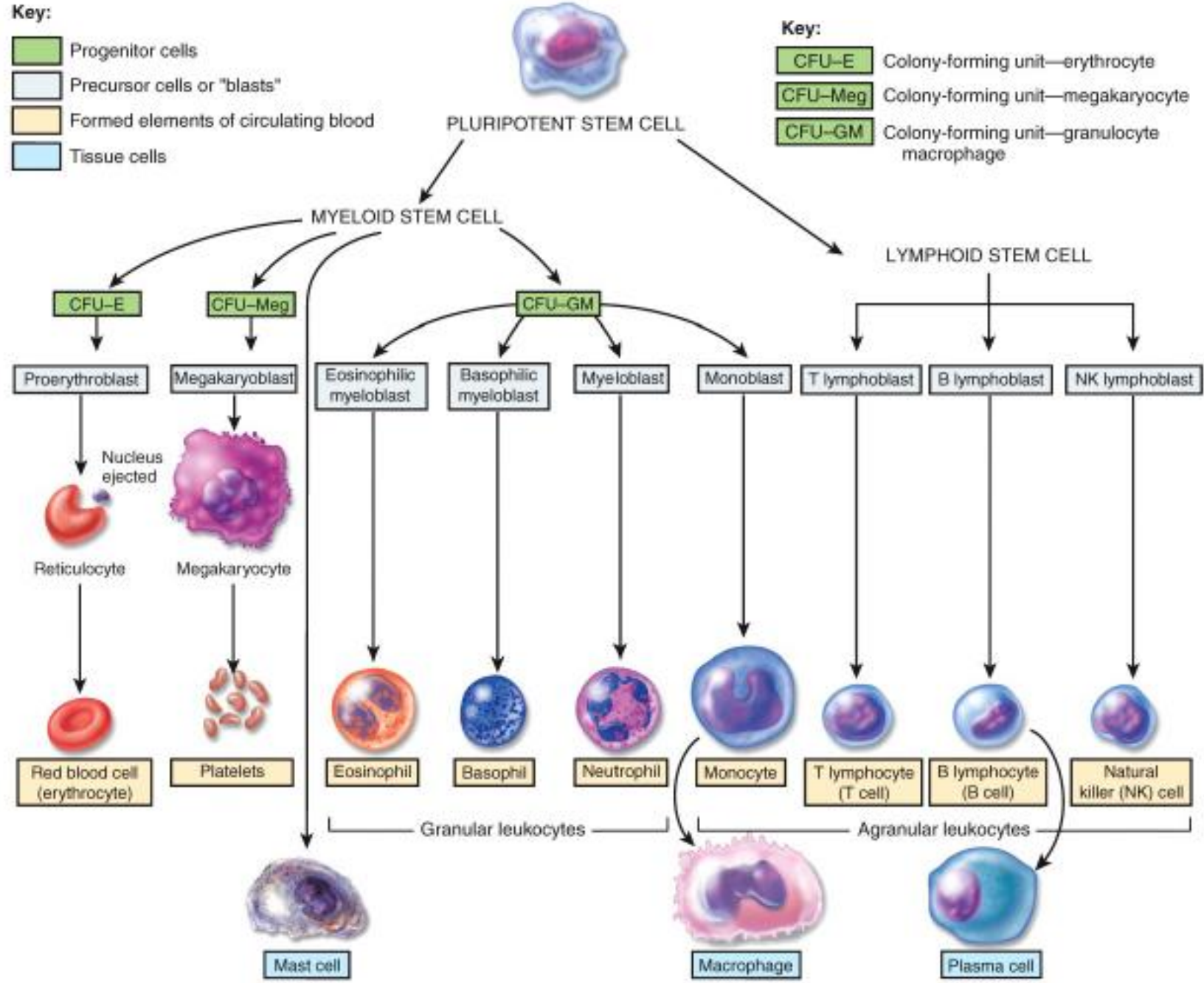


Key:

- Progenitor cells
- Precursor cells or "blasts"
- Formed elements of circulating blood
- Tissue cells

Key:

- CFU-E Colony-forming unit—erythrocyte
- CFU-Meg Colony-forming unit—megakaryocyte
- CFU-GM Colony-forming unit—granulocyte macrophage

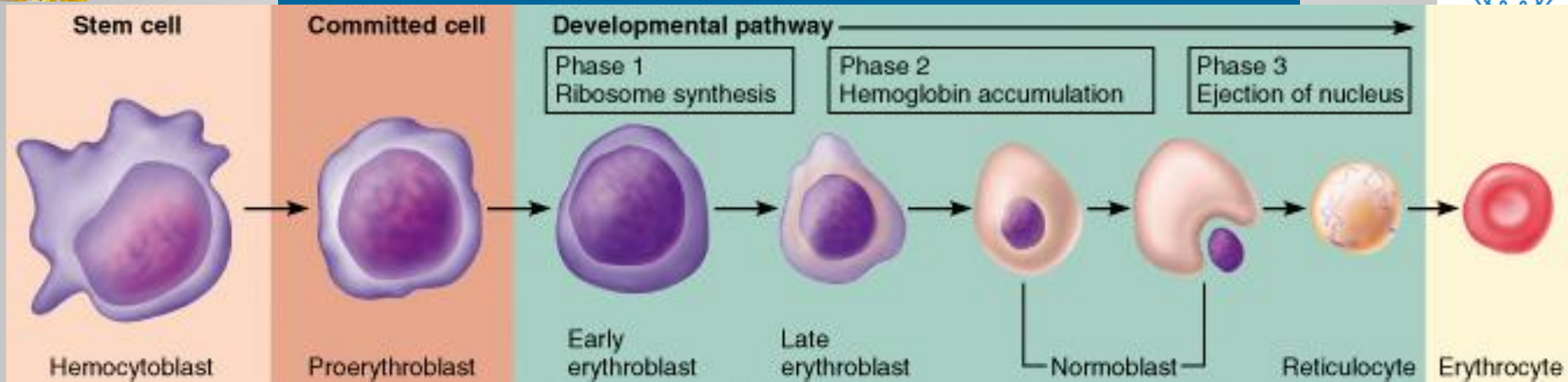


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Factor	Abbreviation	Molecular weight (kDa)	Target cells	Actions
Granulocyte colony-stimulating factor	G-CSF	18	Granulocyte progenitors, mature neutrophils	Increase neutrophil counts
Granulocyte-macrophage colony-stimulating factor	GM-CSF	14–35	Granulocyte, macrophage progenitors, eosinophil progenitors	Increase neutrophil, eosinophil, and monocyte counts
Erythropoietin	EPO	34–39	Erythroid progenitors	Increase red blood cell counts
Stem cell factor	SCF		Granulocyte-erythroid progenitors, lymphoid progenitors, natural killer cells	Increase pluripotent stem cells and progenitor cells for all other cell types
Thrombopoietin	TPO	35	Stem cells, megakaryocytes, erythroid progenitors	Increase platelet counts
Interleukin-11	IL-11	23	Early hematopoietic progenitors, megakaryocytes	Increase platelet counts

Table 1 ■ Hematopoietic growth factors and their activities.

ERYTHROCYTE



➤ Erythropoiesis

- controlled by hormones, especially **erythropoietin (EPO)** from the kidney
- three phases of RBC maturation
 - production of ribosomes
 - synthesis of hemoglobin
 - ejection of the nucleus and reduction in organelles

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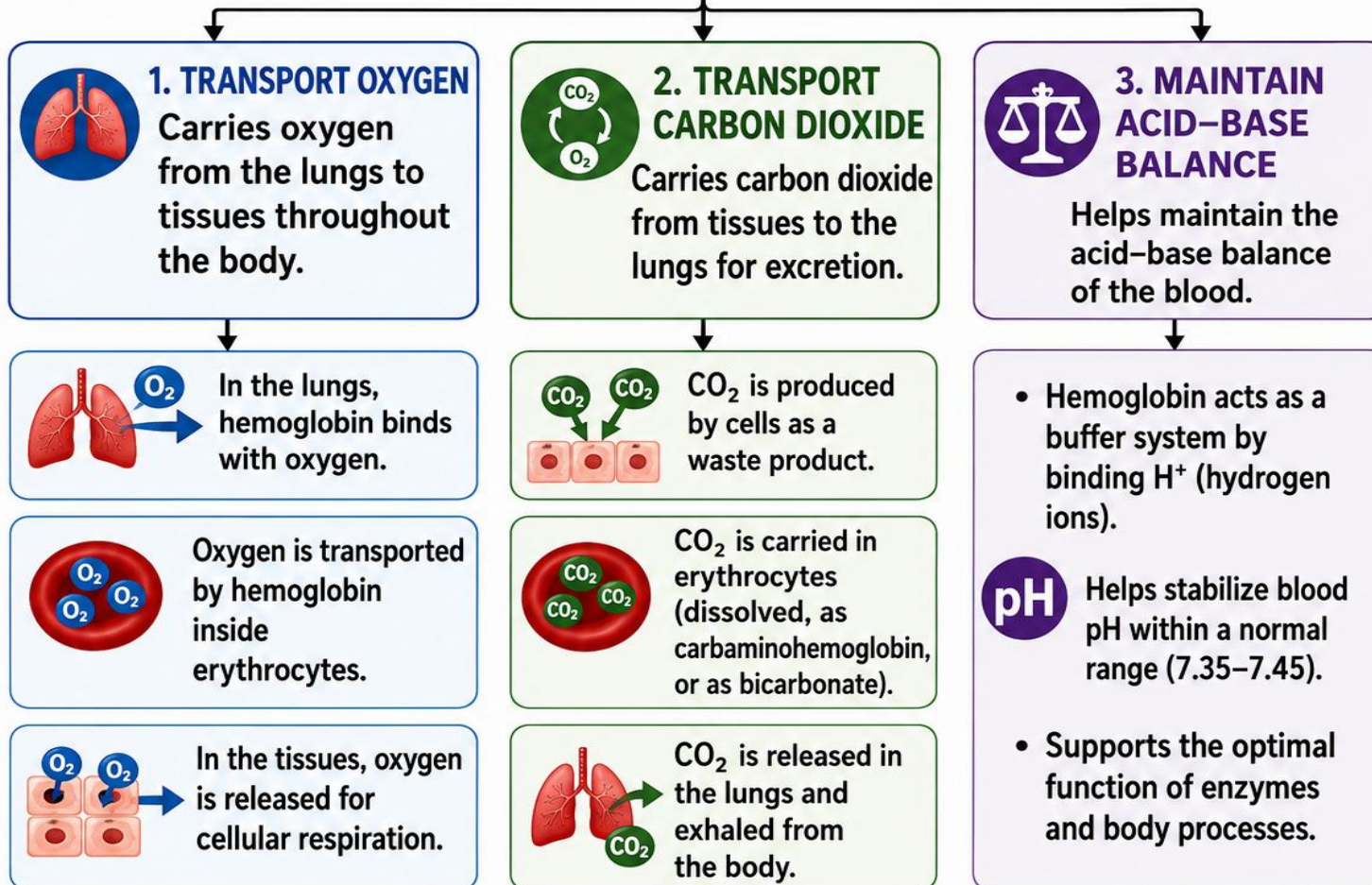
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FUNCTIONS OF ERYTHROCYTES (RED BLOOD CELLS)

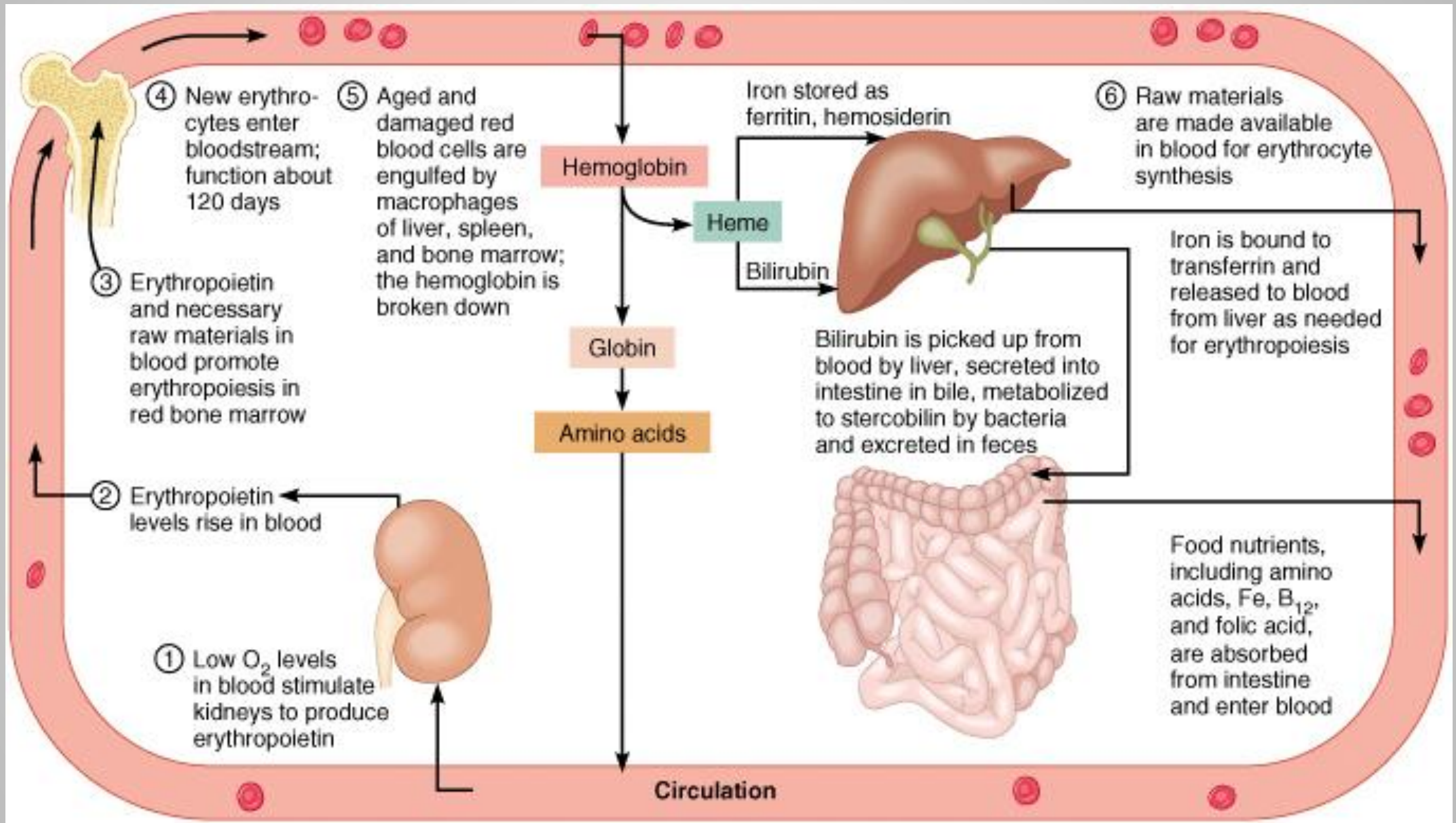


Erythrocytes are specialized cells that contain hemoglobin and circulate in the blood.

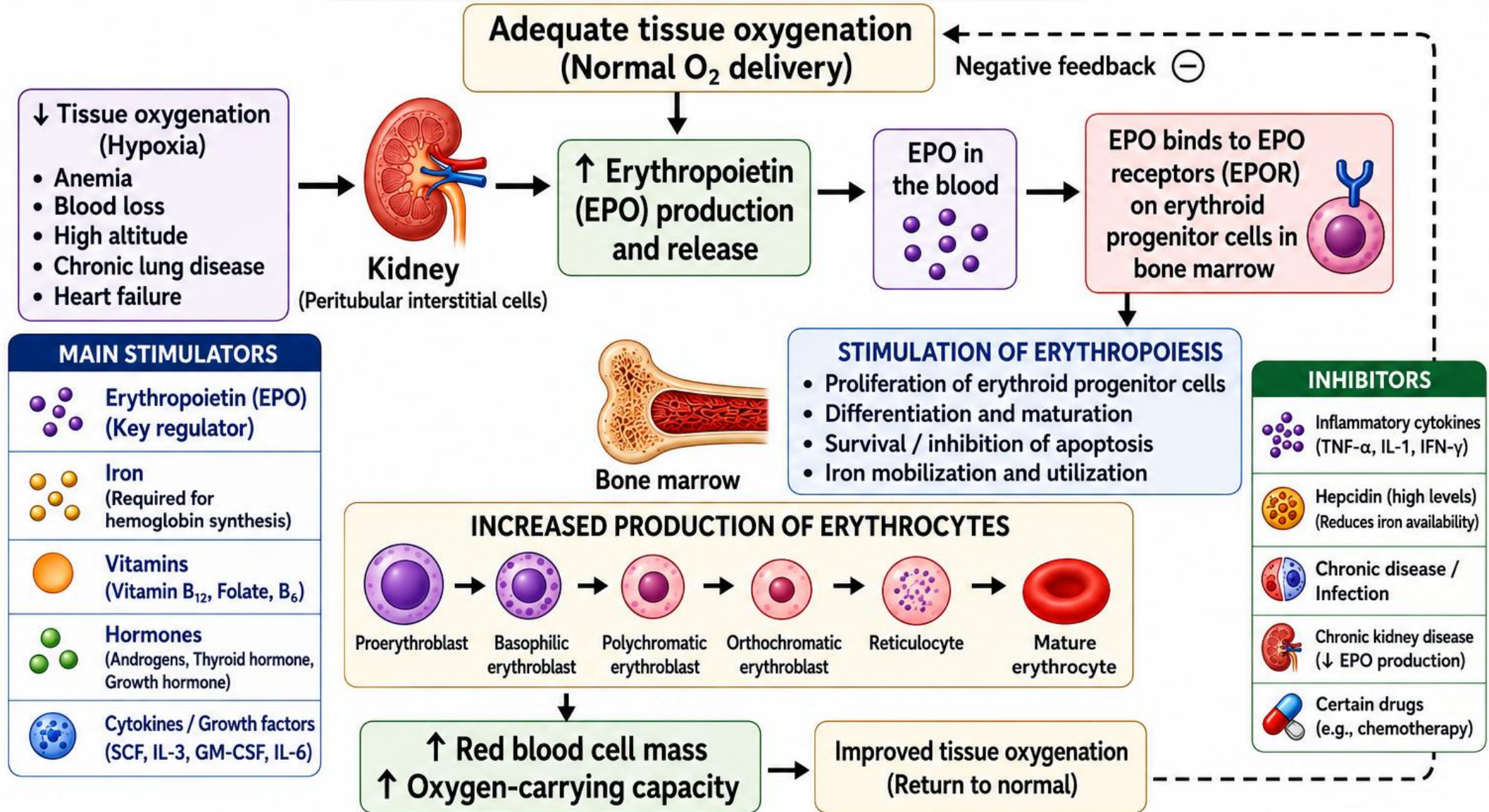
FUNCTIONS OF ERYTHROCYTES



RBC Life Span



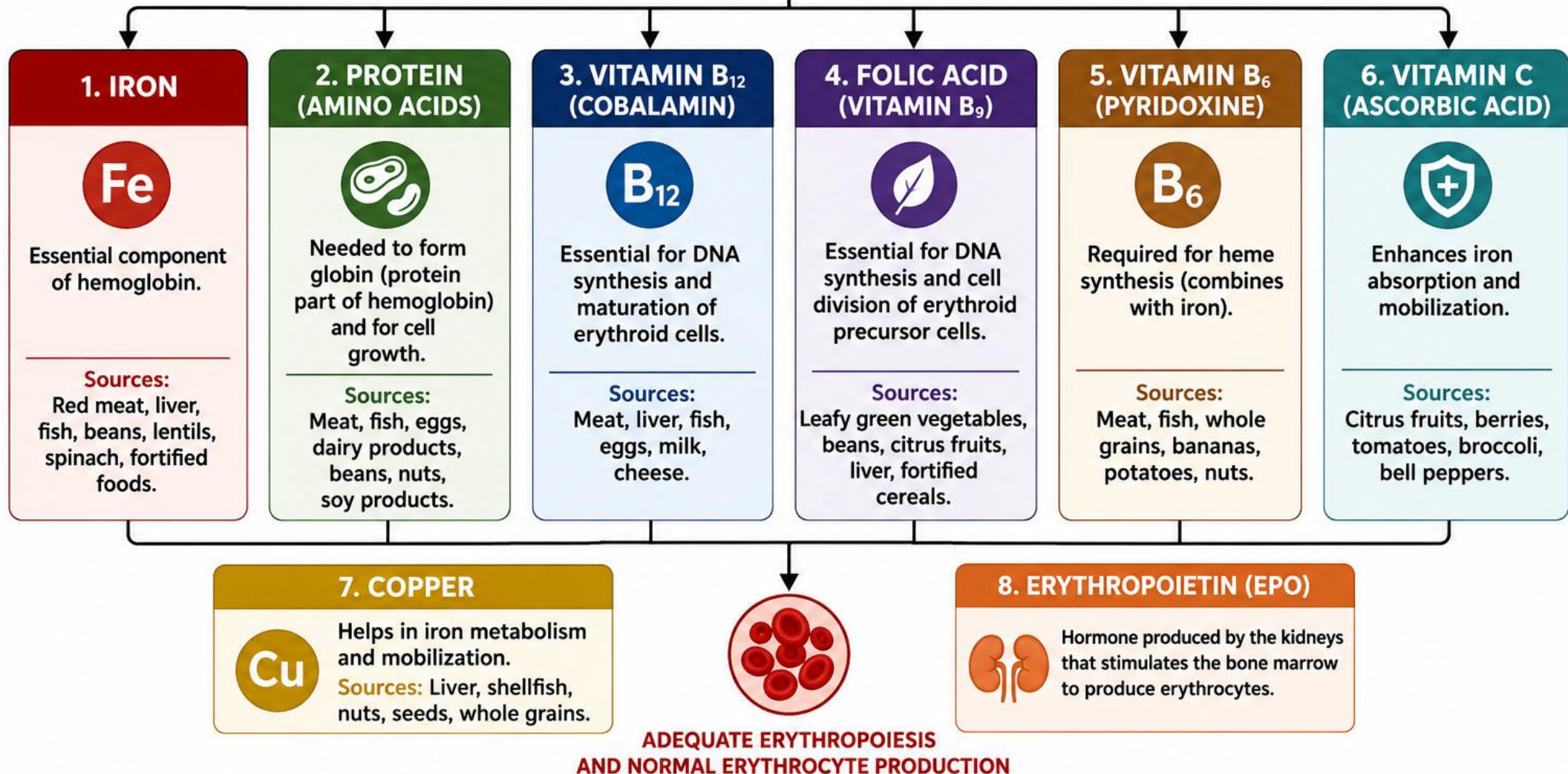
REGULATION OF ERYTHROPOIESIS

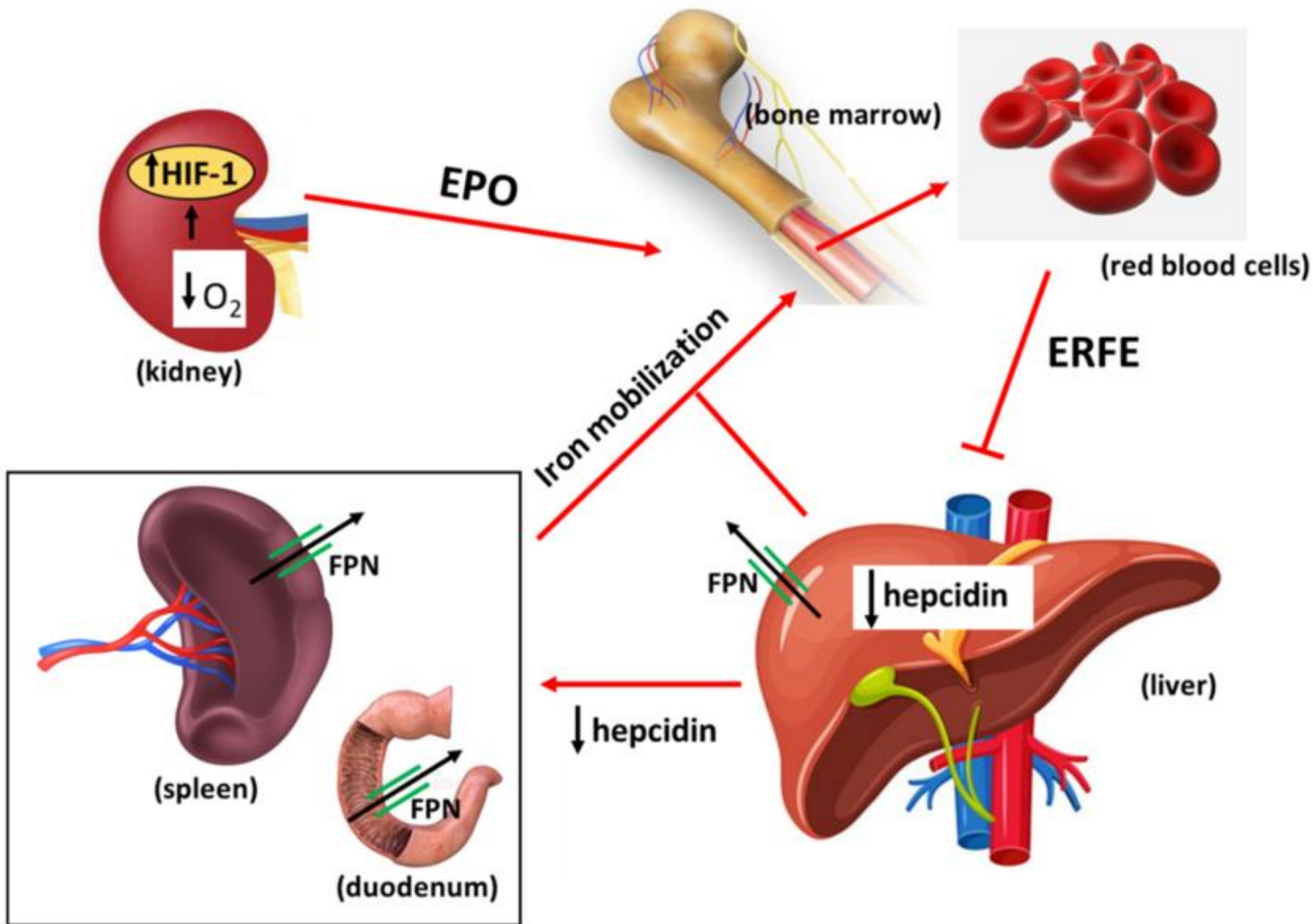


SUMMARY: When tissue oxygen decreases, the kidney releases EPO, which stimulates the bone marrow to produce more erythrocytes. Increased red blood cell mass improves oxygen delivery and reduces EPO production (negative feedback).

NECESSARY SUBSTANCES FOR ERYTHROPOIESIS

Erythropoiesis is the formation of erythrocytes in the bone marrow.





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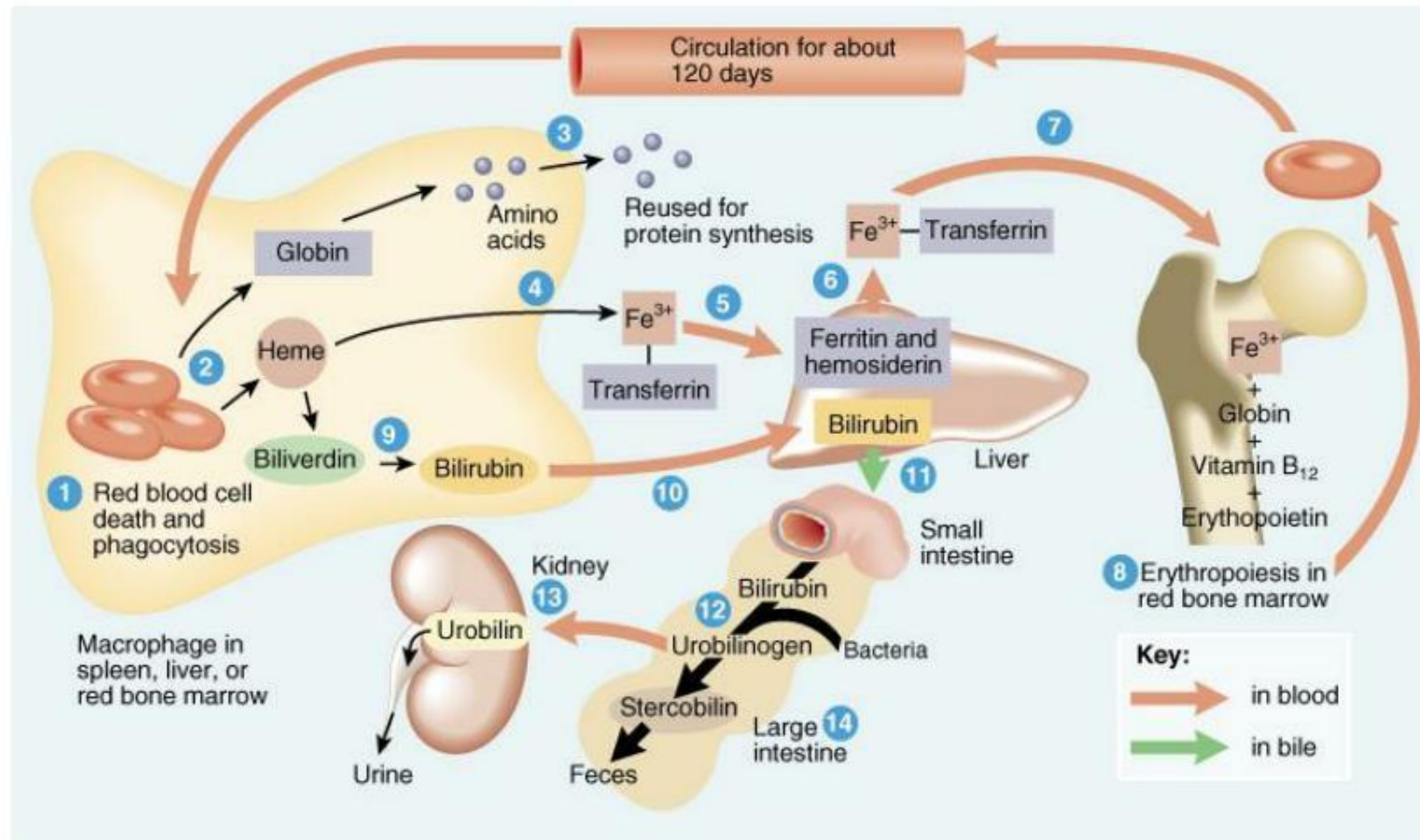


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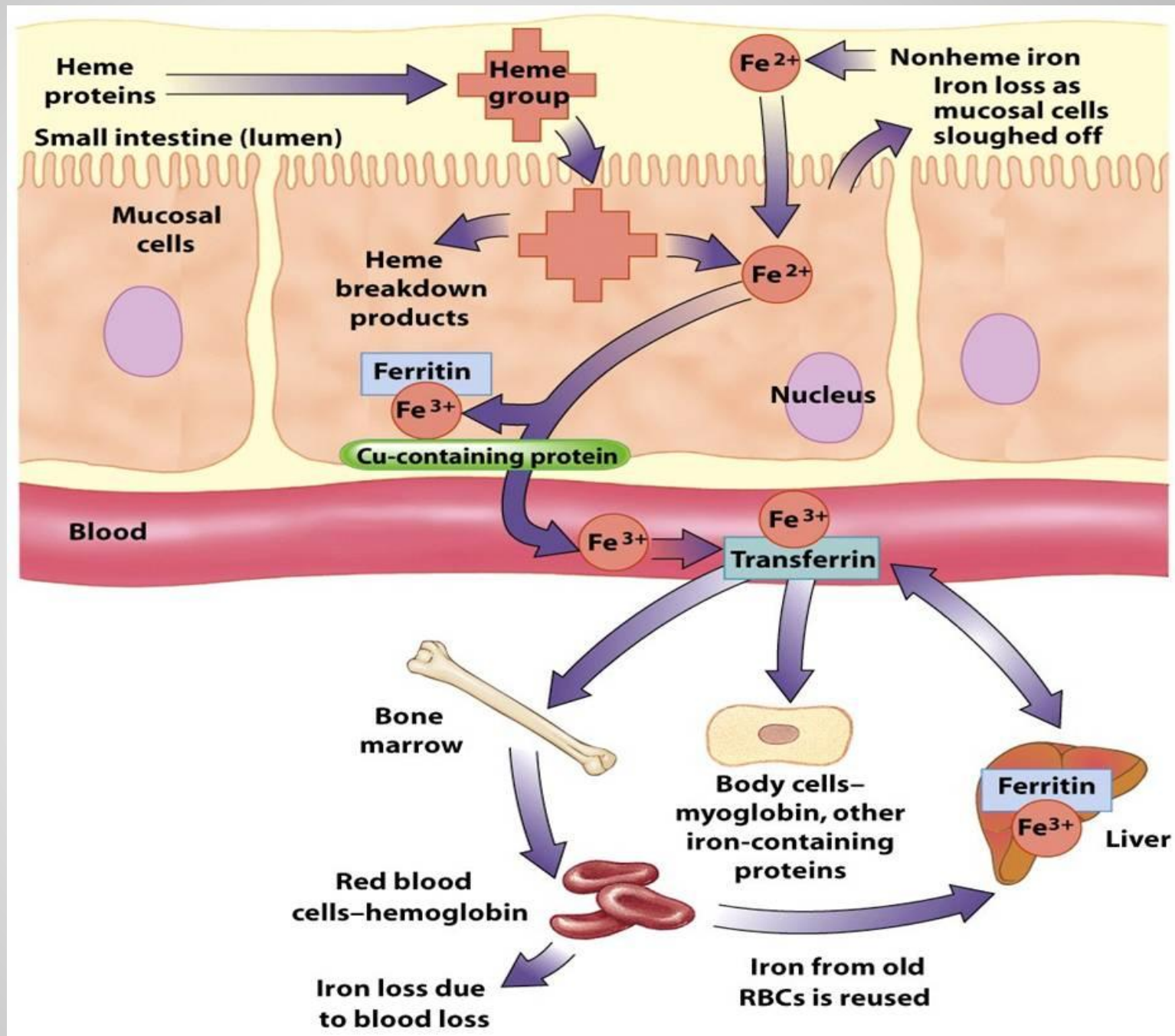
Recycling of Hemoglobin



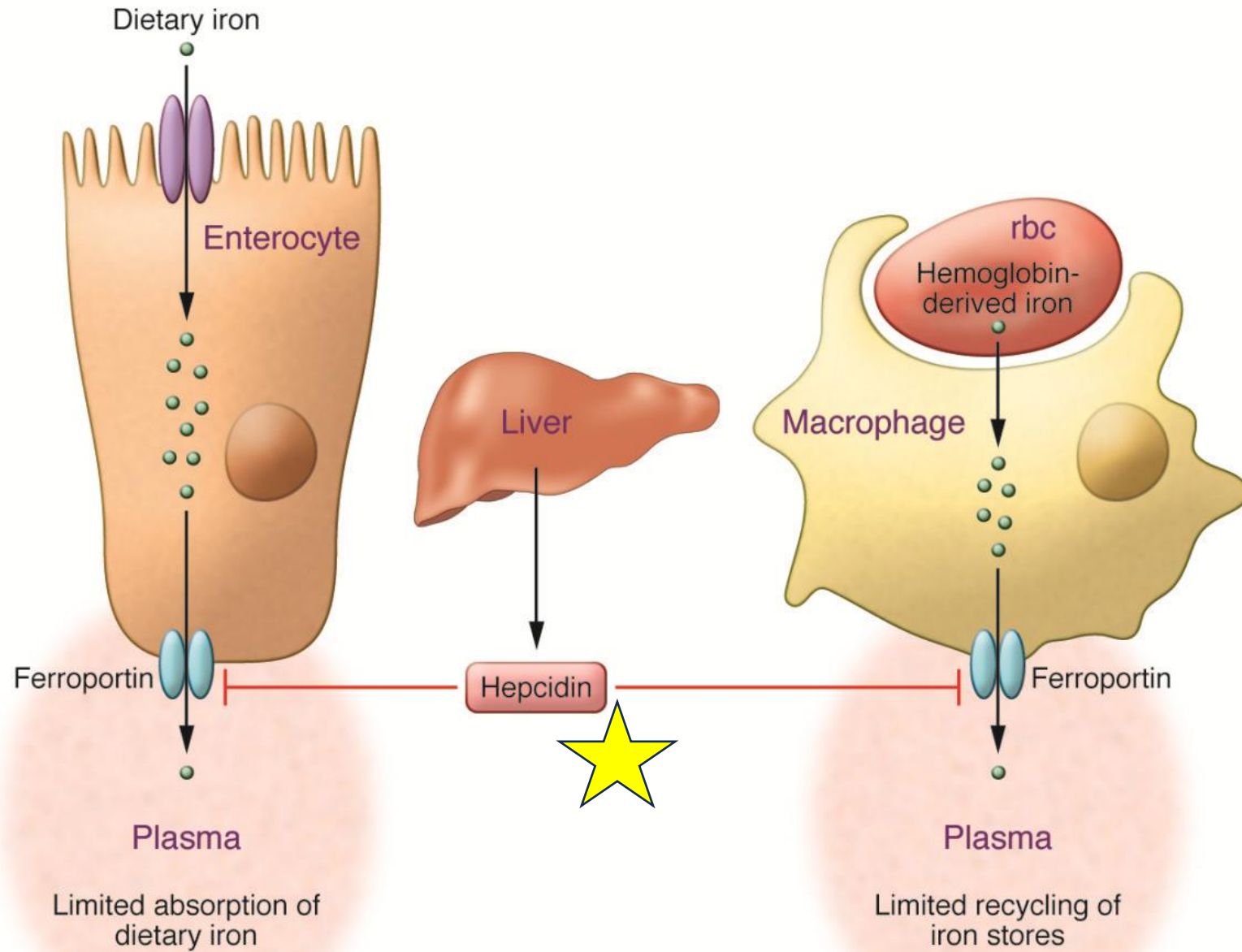
In macrophages of liver or spleen

- globin portion broken down into amino acids & recycled
- heme portion split into iron (Fe^{3+}) and biliverdin (green pigment)

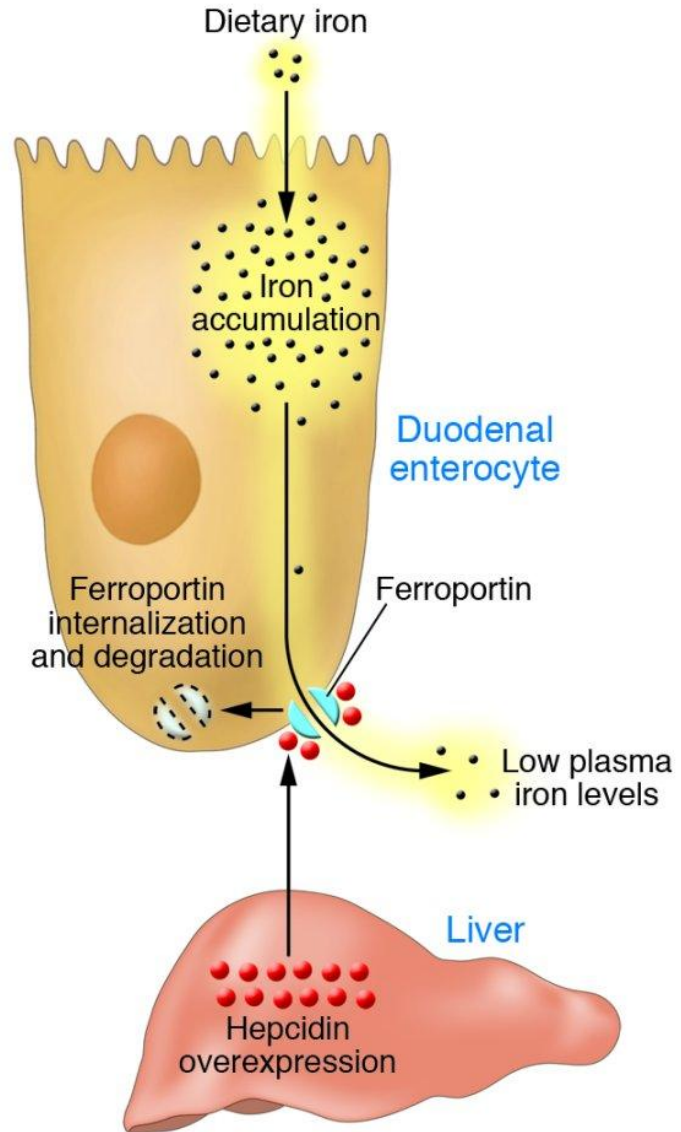
Iron transport



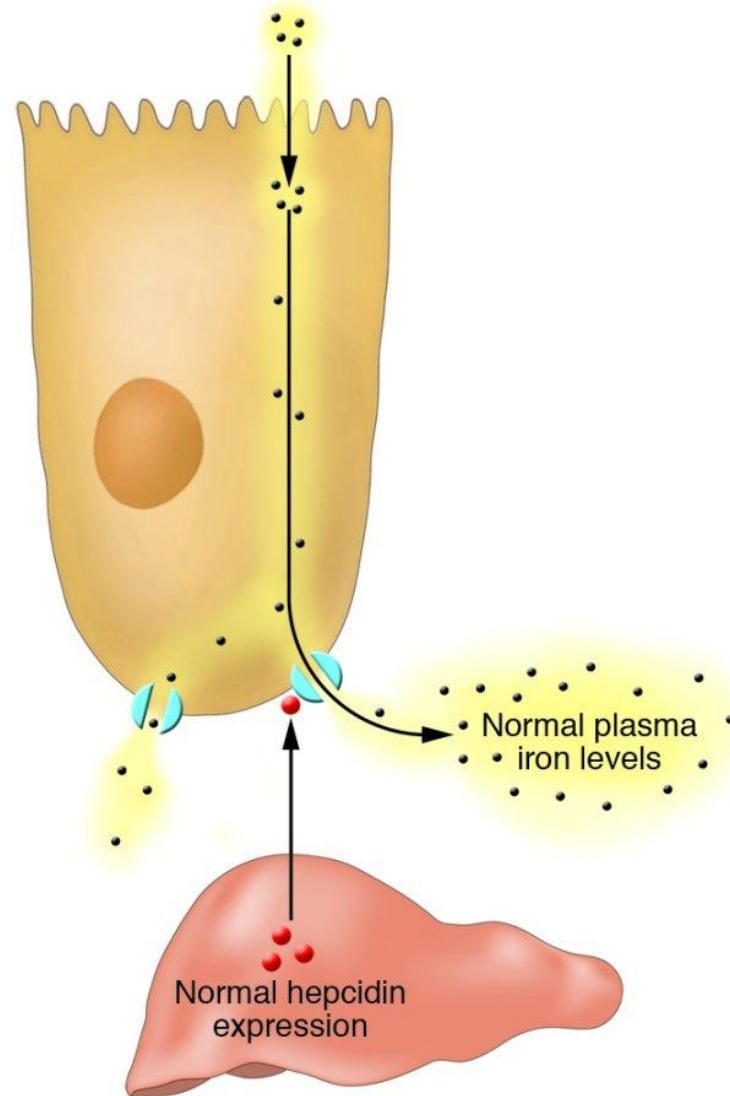
Iron transport regulation



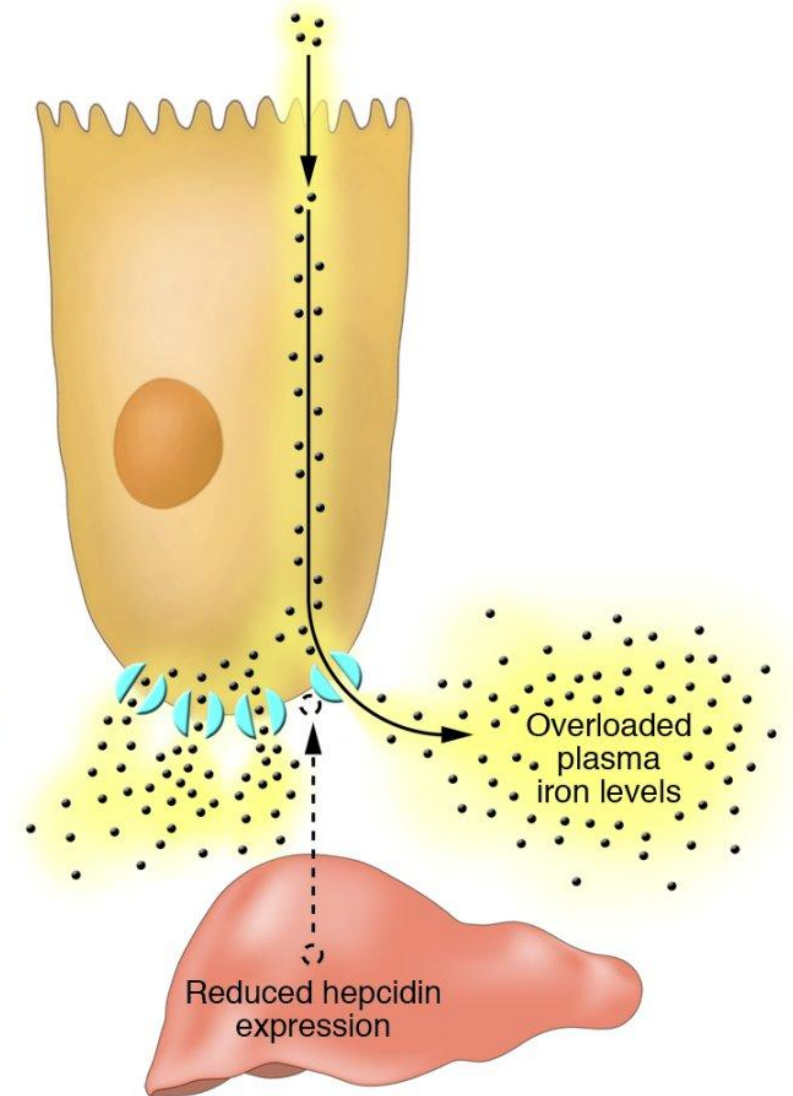
A Anemia of chronic disease



B Normal

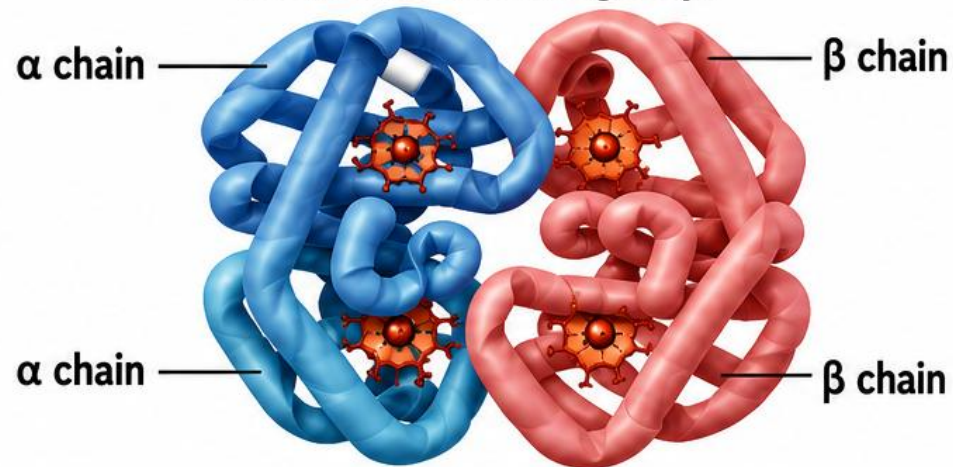


C Hemochromatosis (iron overload)



STRUCTURE OF HEMOGLOBIN

Hemoglobin is a tetrameric protein composed of four polypeptide chains, each with a heme group.



Each polypeptide chain binds one heme group, and each heme group binds one iron (Fe^{2+}) atom.

1 GLOBIN (PROTEIN PART)

- Hemoglobin is made of 4 polypeptide chains (globin chains).
- In adult hemoglobin (HbA), the composition is: $\alpha_2\beta_2$ (2 alpha chains and 2 beta chains)



2 HEME (NON-PROTEIN PART)

- Each globin chain has one heme group.
- Heme is responsible for binding oxygen.



3 IRON (Fe^{2+})

- Ferrous iron (Fe^{2+}) is located in the center of the heme group.
- It binds reversibly with oxygen (O_2).



GLOBIN SUBUNITS

Subunit	Polypeptide Chain
α (alpha)	141 amino acids
β (beta)	146 amino acids
Total	4 chains ($\alpha_2\beta_2$)

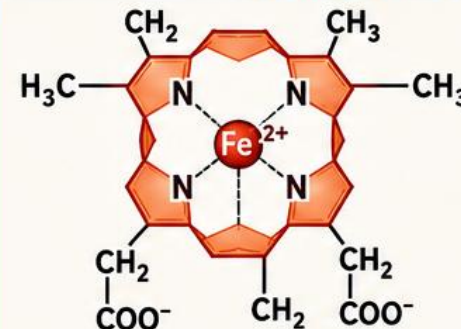
COMPOSITION OF HEMOGLOBIN (HbA)

- 2 α chains
- 2 β chains
- 4 heme groups
- 4 iron atoms (Fe^{2+})



Molecular weight: ~64,500 Da

STRUCTURE OF HEME



- Heme is a porphyrin ring with four nitrogen atoms that coordinate the iron (Fe^{2+}) in the center.
- Each heme binds one O_2 molecule.



Types of adult hemoglobin

Form	Chain composition	Fraction of total hemoglobin
HbA	$\alpha_2\beta_2$	90%
HbF	$\alpha_2\gamma_2$	<2%
HbA ₂	$\alpha_2\delta_2$	2–5%
HbA _{1c}	$\alpha_2\beta_2$ -glucose	< 5 %

HbA1: the **major** hemoglobin in humans

HbA2: first appears 12 weeks after birth

- a **minor** component of normal adult HB increased in ***β-thalassemia***

HbF: normally synthesized only during **fetal** development
in fetus and newborn infants Hb F binds O₂ at lower tension than Hb A
→ Hb F has a higher affinity to O₂
After birth, Hb F is replaced by Hb A during the first few months of life.

HbA1C: has **glucose** residues attached to β-globin chains
– increased amounts in **DM**

NORMAL VALUES OF ERYTHROCYTES AND HEMOGLOBIN AT VARIOUS AGES ACCORDING TO WHO

(WHO, 2011 – Haemoglobin concentrations for the diagnosis of anaemia and assessment of severity)

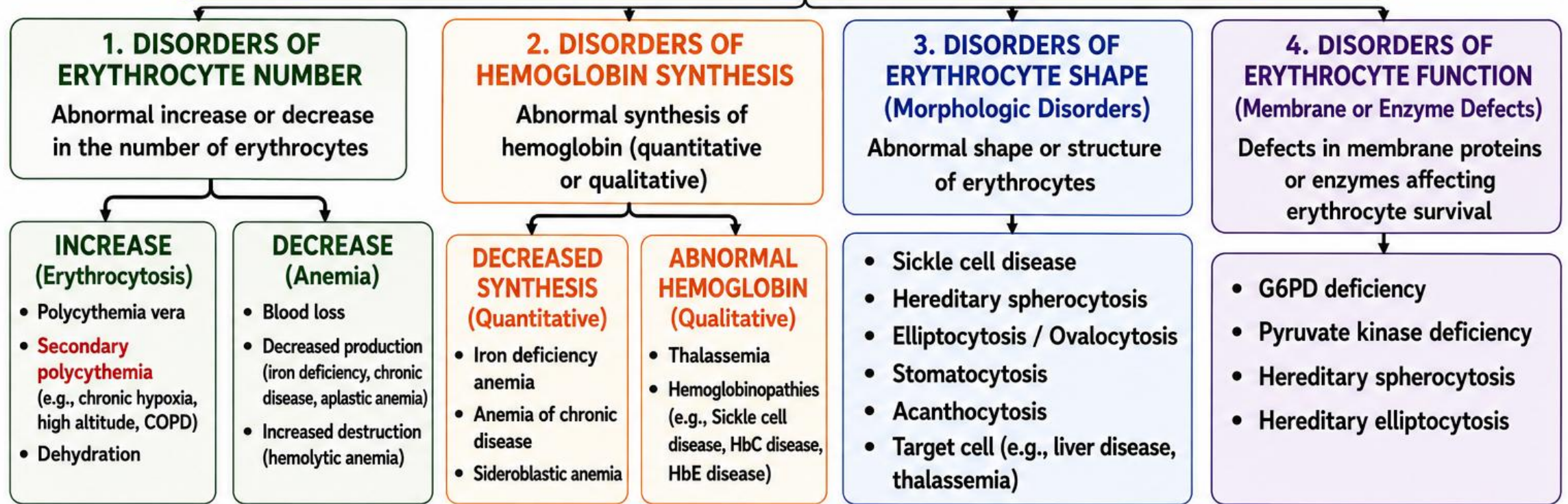
AGE GROUP		ERYTHROCYTES (RBC)			HEMOGLOBIN (Hb)		
		Normal Range (×10 ¹² cells/L)		Unit	Normal Range (g/dL)		Unit
	Newborn (0 – 1 month)	4.0 – 6.6		×10 ¹² cells/L	13.5 – 24.0		g/dL
	Infants (1 – 12 months)	3.6 – 5.2		×10 ¹² cells/L	10.5 – 13.5		g/dL
	Children (1 – 5 years)	3.9 – 5.3		×10 ¹² cells/L	11.0 – 13.0		g/dL
	Children (5 – 11 years)	4.0 – 5.5		×10 ¹² cells/L	11.5 – 15.5		g/dL
	Adolescents (12 – 14 years)	4.2 – 5.6		×10 ¹² cells/L	12.0 – 16.0		g/dL
	Adolescents (15 – 17 years)	Male: 4.3 – 5.9	Female: 3.9 – 5.1	×10 ¹² cells/L	Male: 13.0 – 17.0	Female: 12.0 – 15.0	g/dL
	Adults (≥ 18 years)	Male: 4.5 – 5.9	Female: 4.0 – 5.2	×10 ¹² cells/L	Male: 13.0 – 17.0	Female: 12.0 – 15.0	g/dL

★ Notes:

- RBC (Erythrocytes) count is expressed in $\times 10^{12}$ cells/L (teraliters).
- Hemoglobin (Hb) concentration is expressed in g/dL (grams per deciliter).
- These values are reference ranges for healthy individuals living at sea level.

ERYTHROCYTE DISORDERS

(Abnormalities in number, hemoglobin, size, shape, or function)



EXAMPLES OF ABNORMAL ERYTHROCYTE MORPHOLOGY

	Normal (biconcave disc)		Sickle cell (falciform)		Spherocyte (small, no central pallor)		Elliptocyte / Ovalocyte (elliptical shape)
	Target cell (target appearance)		Stomatocyte (mouth-shaped central pallor)		Acanthocyte (spiky projections)		Schistocyte (fragmented cell)



Anemia



- Causes of anemia :
 - **inadequate erythropoiesis or hemoglobin synthesis**
 - kidney failure and insufficient erythropoietin
 - iron-deficiency anemia
 - inadequate vitamin B₁₂ from poor nutrition or lack of intrinsic factor (pernicious anemia)
 - aplastic anemia - complete cessation of erythropoiesis
 - **hemorrhagic anemias from bleeding**
 - **hemolytic anemias from RBC destruction (hemolytic)**

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Erythrocyte Disorders



Polycythemia - an excess of RBCs

- Primary polycythemia (polycythemia vera)
 - cancer of erythropoietic cell line in red bone marrow
 - RBC count as high as 11 million/ μ L; hematocrit 80%
- Secondary polycythemia
 - from dehydration, emphysema, high altitude, or physical conditioning
 - RBC count up to 8 million/ μ L

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Thrombocytes (PLATELETS)

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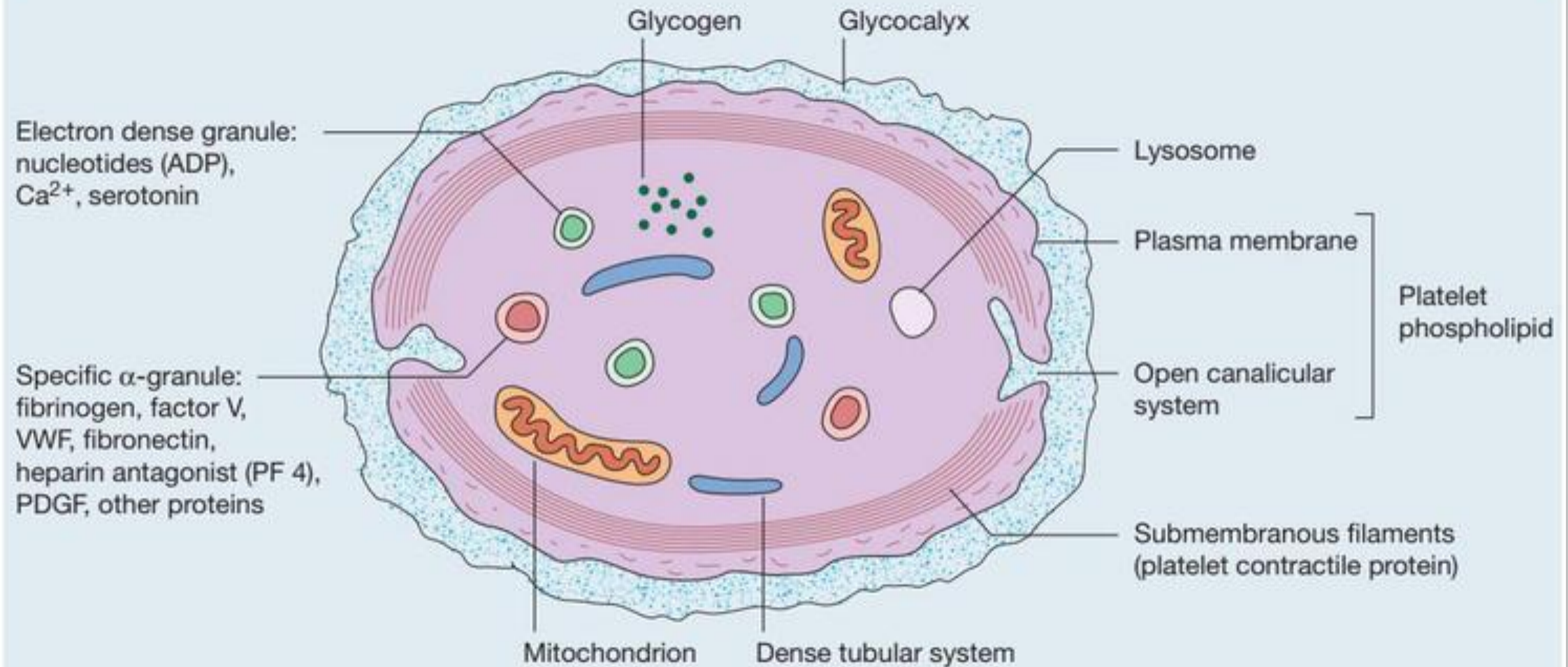
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Thrombocytes Structure



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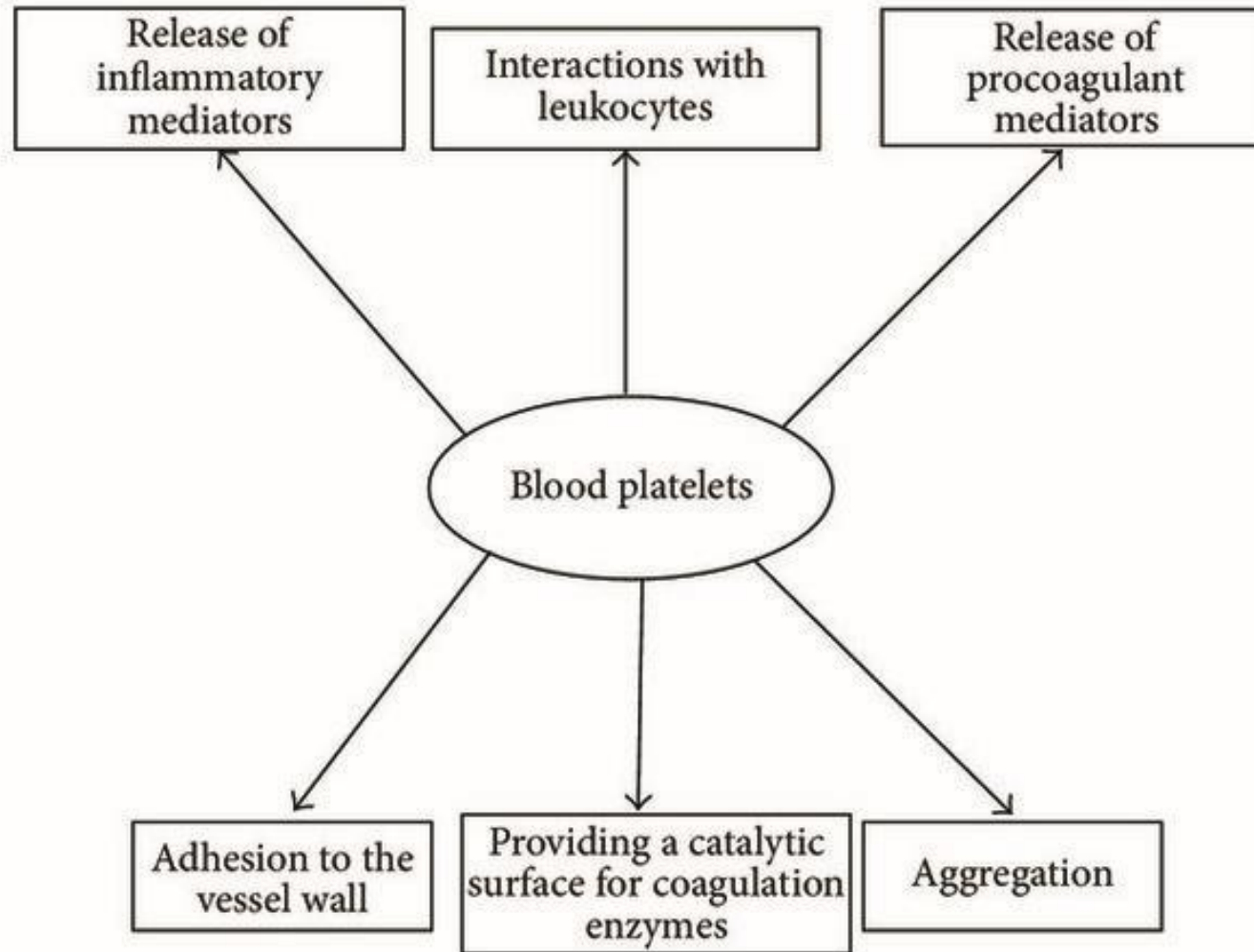


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Function of Thrombocyte



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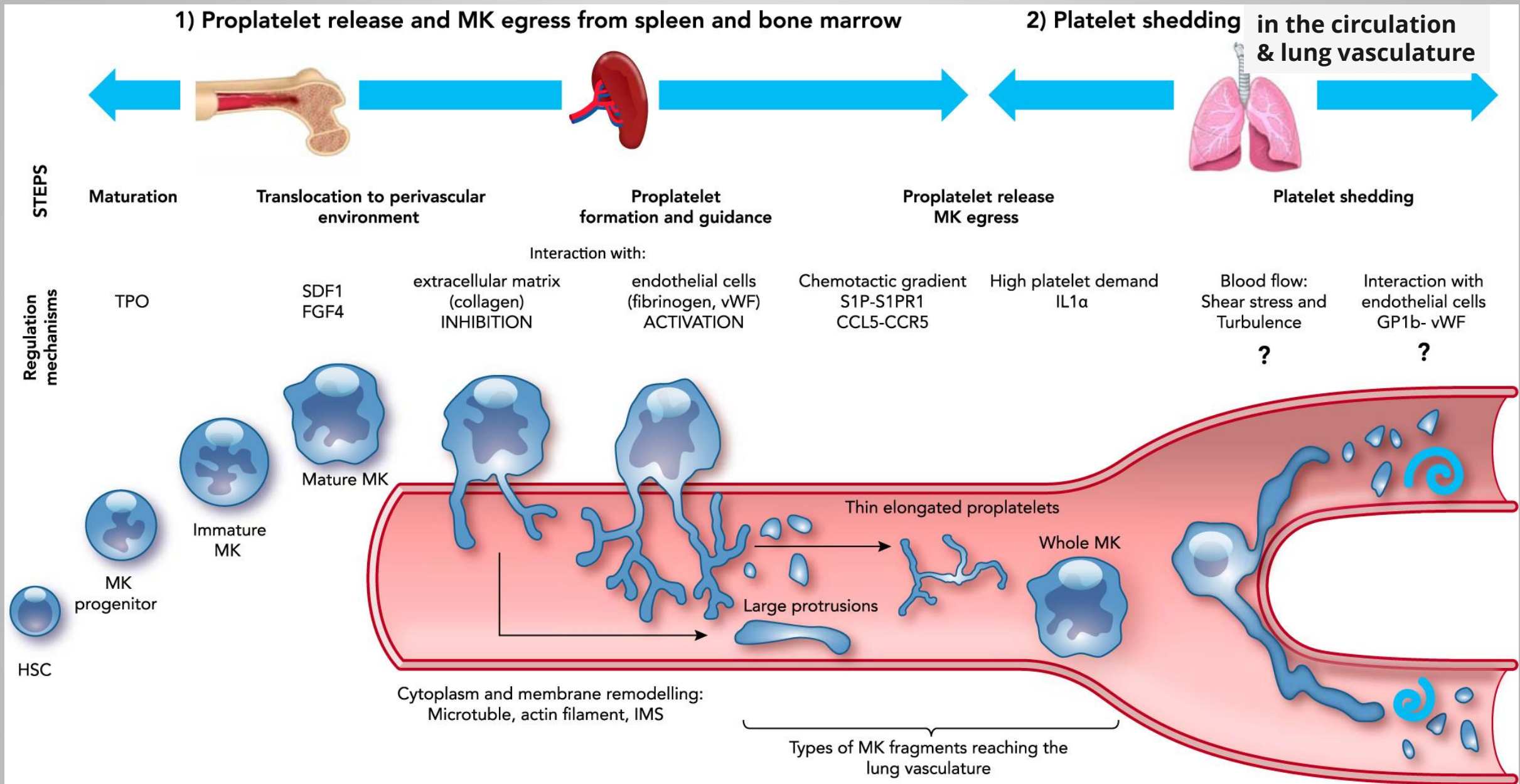
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Direct contact with blood vessels, large cytoplasm, and expression of mature MK signatures are important factors for MKs to optimally produce platelets.

NORMAL PLATELET COUNT

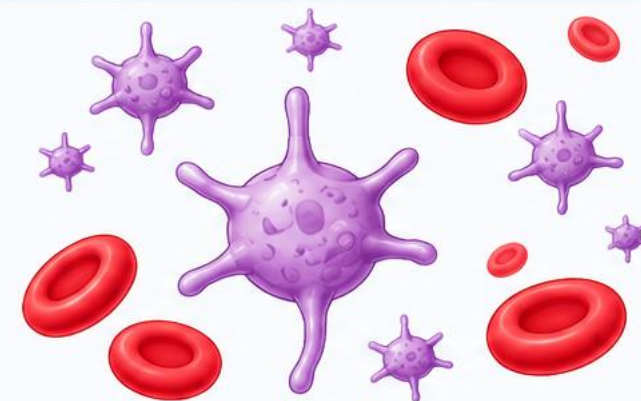
Platelets (thrombocytes) are small cell fragments that play a crucial role in hemostasis and prevention of bleeding.

NORMAL PLATELET COUNT RANGE

150,000 – 450,000

platelets per microliter (μL) of blood

($150 \times 10^9 - 450 \times 10^9$ platelets/L)



PLATELET COUNT CLASSIFICATION	PLATELET COUNT (platelets/ μL)
NORMAL	150,000 – 450,000
THROMBOCYTOPENIA (Low platelet count)	< 150,000
THROMBOCYTOSIS (High platelet count)	> 450,000

IMPORTANT NOTES



The normal platelet count may vary slightly between laboratories.



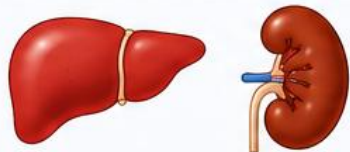
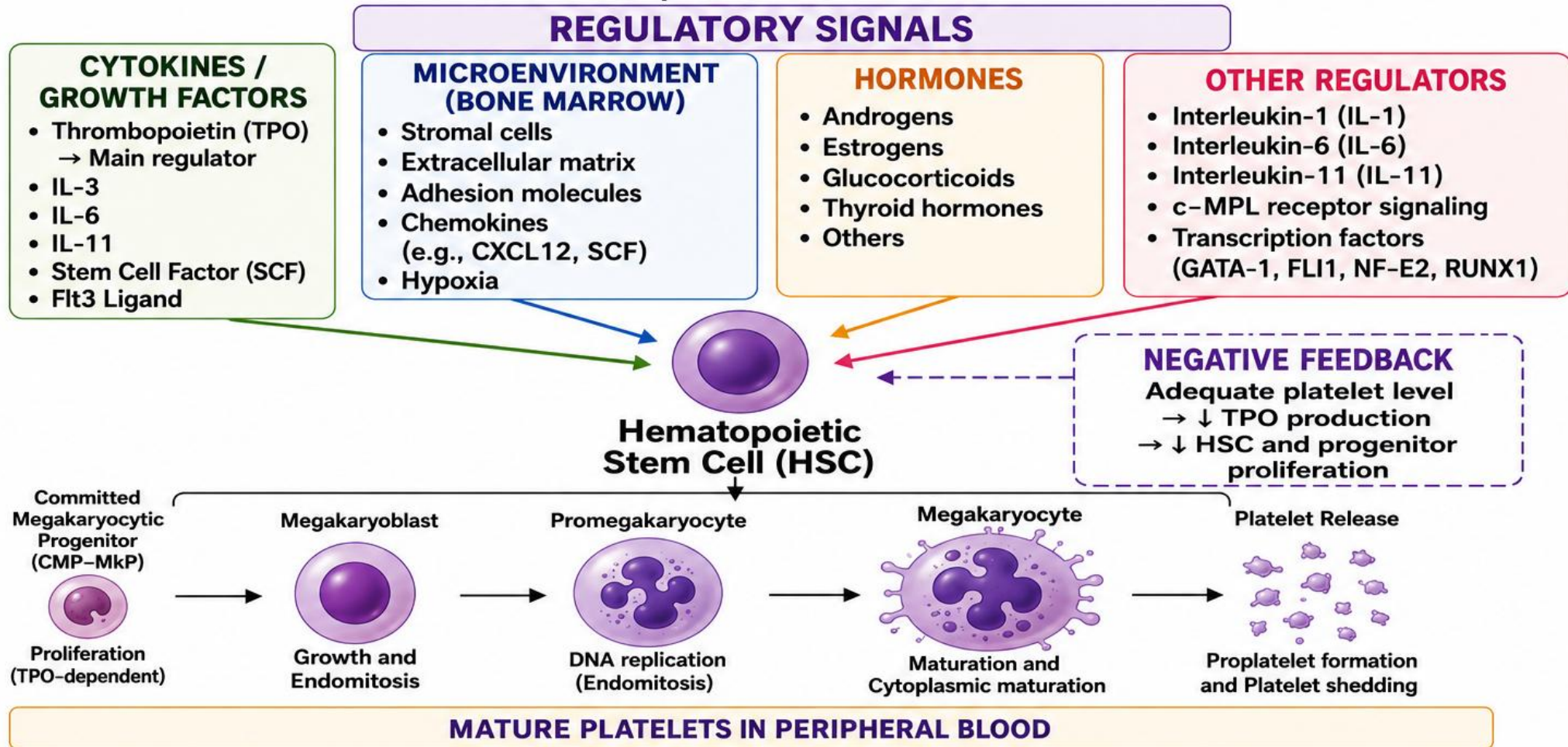
Platelet count is an important part of a complete blood count (CBC) and helps evaluate bleeding and clotting disorders.



If your platelet count is outside the normal range, further evaluation by a healthcare professional is recommended.

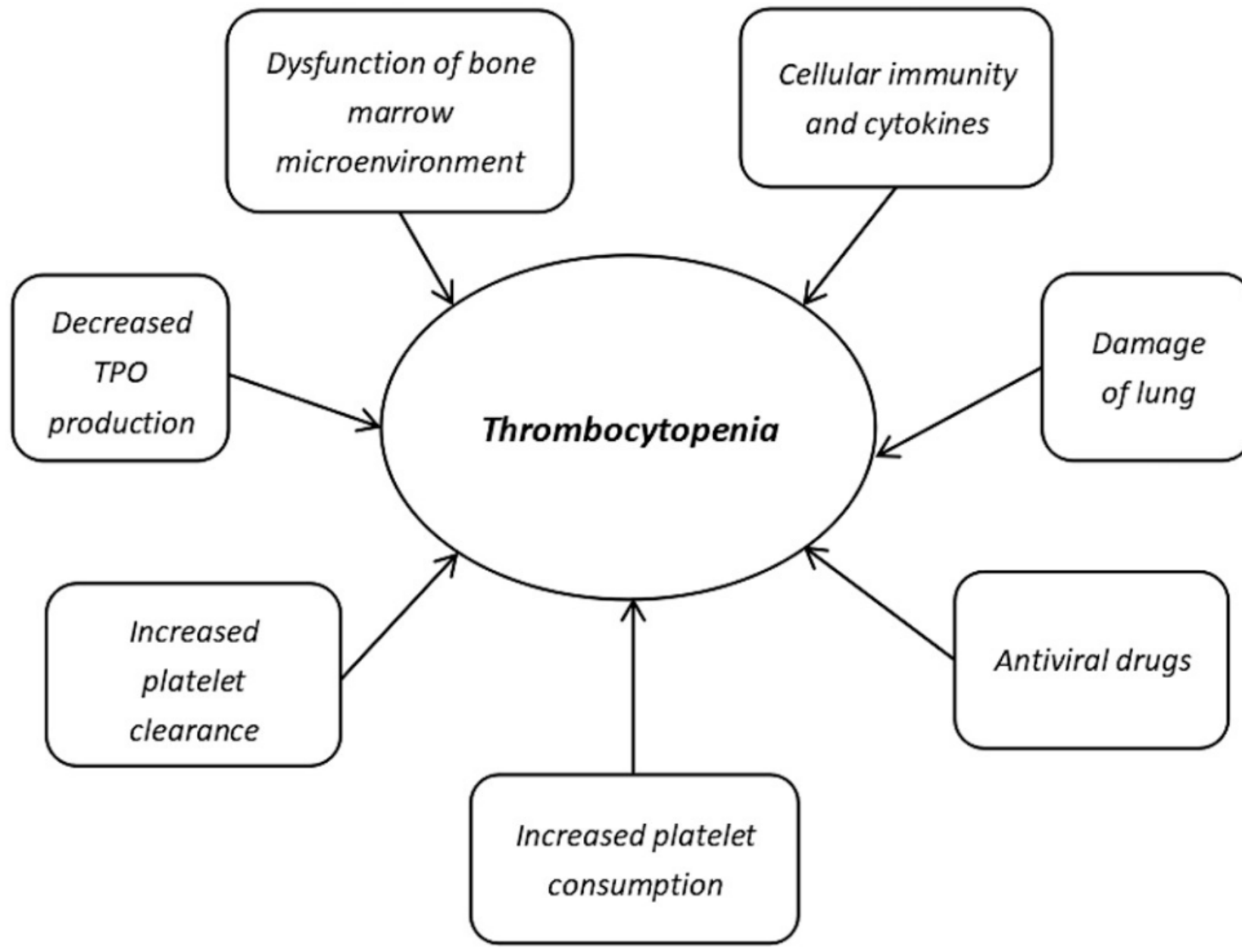
REGULATION OF THROMBOPOIESIS

Thrombopoiesis is the process of platelet production in the bone marrow, tightly regulated by thrombopoietin (TPO) and other factors to maintain platelet homeostasis.



THROMBOPOIETIN (TPO)

- Mainly produced by the liver (hepatocytes) and kidneys.
- Circulates in plasma and binds to c-MPL receptor on HSCs and megakaryocytic progenitors.
- TPO level is inversely related to platelet mass (clearance by platelets and megakaryocytes).



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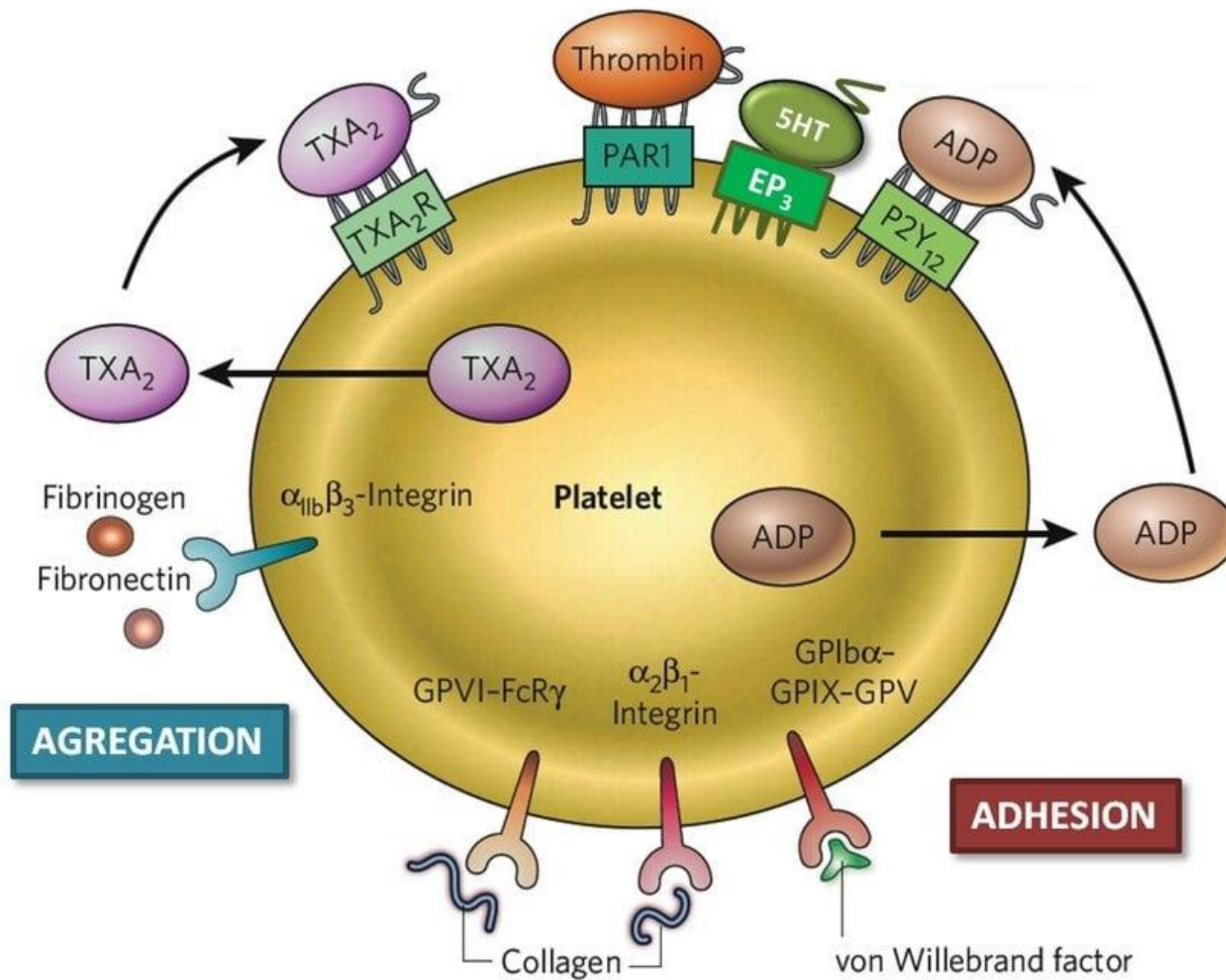
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ACTIVATION

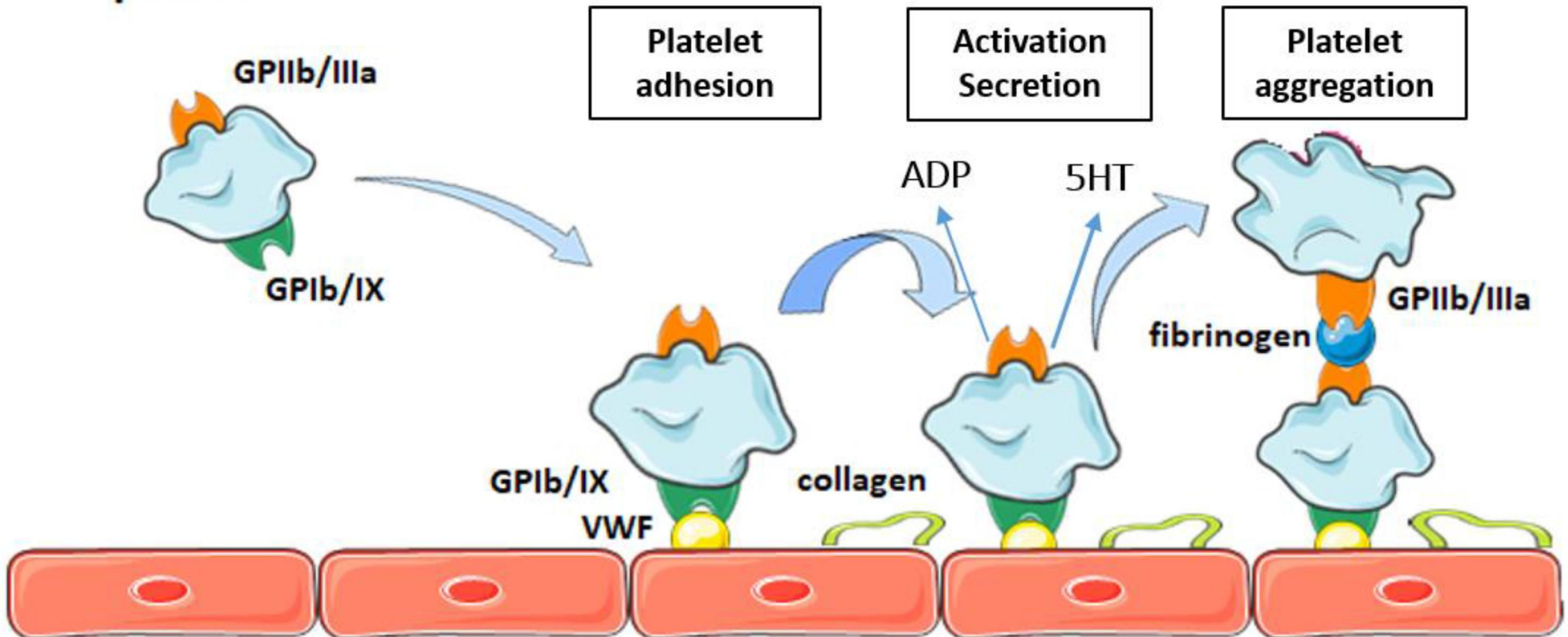


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Endothelium



platelet





LEUKOCYTE

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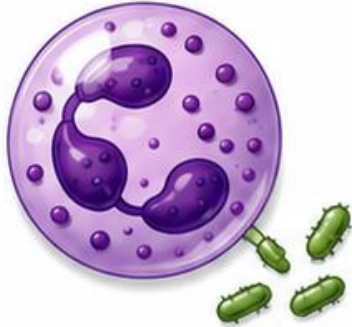


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FUNCTIONS OF LEUKOCYTES

Leukocytes are immune cells that protect the body from infections, remove abnormal cells, and regulate immune responses.

NEUTROPHILS



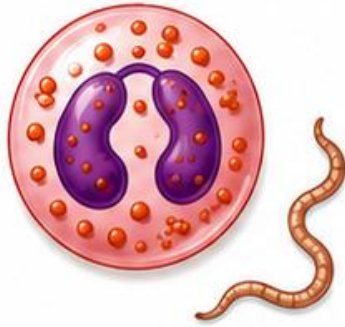
MAIN FUNCTION

Phagocytosis and killing of bacteria and fungi (first responders in acute inflammation).

HOW THEY WORK

Move to the site of infection, engulf pathogens, release enzymes and reactive oxygen species.

EOSINOPHILS



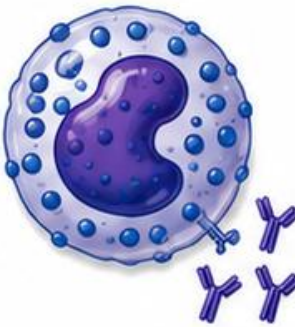
MAIN FUNCTION

Defense against parasites, especially helminths, and modulation of allergic reactions.

HOW THEY WORK

Release cytotoxic granule proteins; modulate inflammation in allergic responses.

BASOPHILS



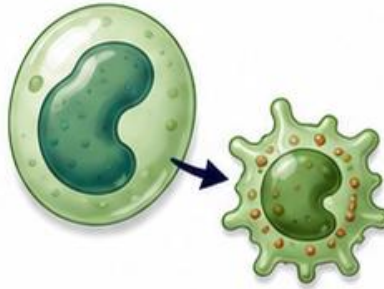
MAIN FUNCTION

Release mediators of inflammation and allergic responses.

HOW THEY WORK

Release histamine, heparin, and other mediators that cause vasodilation and increase vascular permeability.

MONOCYTES



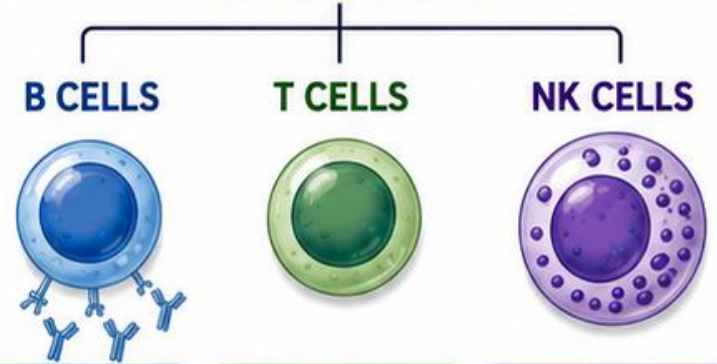
MAIN FUNCTION

Phagocytosis of pathogens, dead cells, and debris; present antigens to lymphocytes.

HOW THEY WORK

Circulate in blood as monocytes and differentiate into macrophages or dendritic cells in tissues.

LYMPHOCYTES



B CELLS



MAIN FUNCTION

Produce antibodies; provide humoral immune response.

T CELLS



MAIN FUNCTION

Cell-mediated immunity; help and regulate other immune cells.

NK CELLS



MAIN FUNCTION

Kill virus-infected cells and tumor cells without prior sensitization.

HOW THEY WORK

Recognize specific antigens and eliminate infected or abnormal cells; regulate and coordinate immune responses.

NORMAL LEUKOCYTE COUNT (According to Adults)

TOTAL LEUKOCYTE COUNT (WBC)

4,000 – 11,000 cells/ μ L
(or $4.0 - 11.0 \times 10^9/L$)

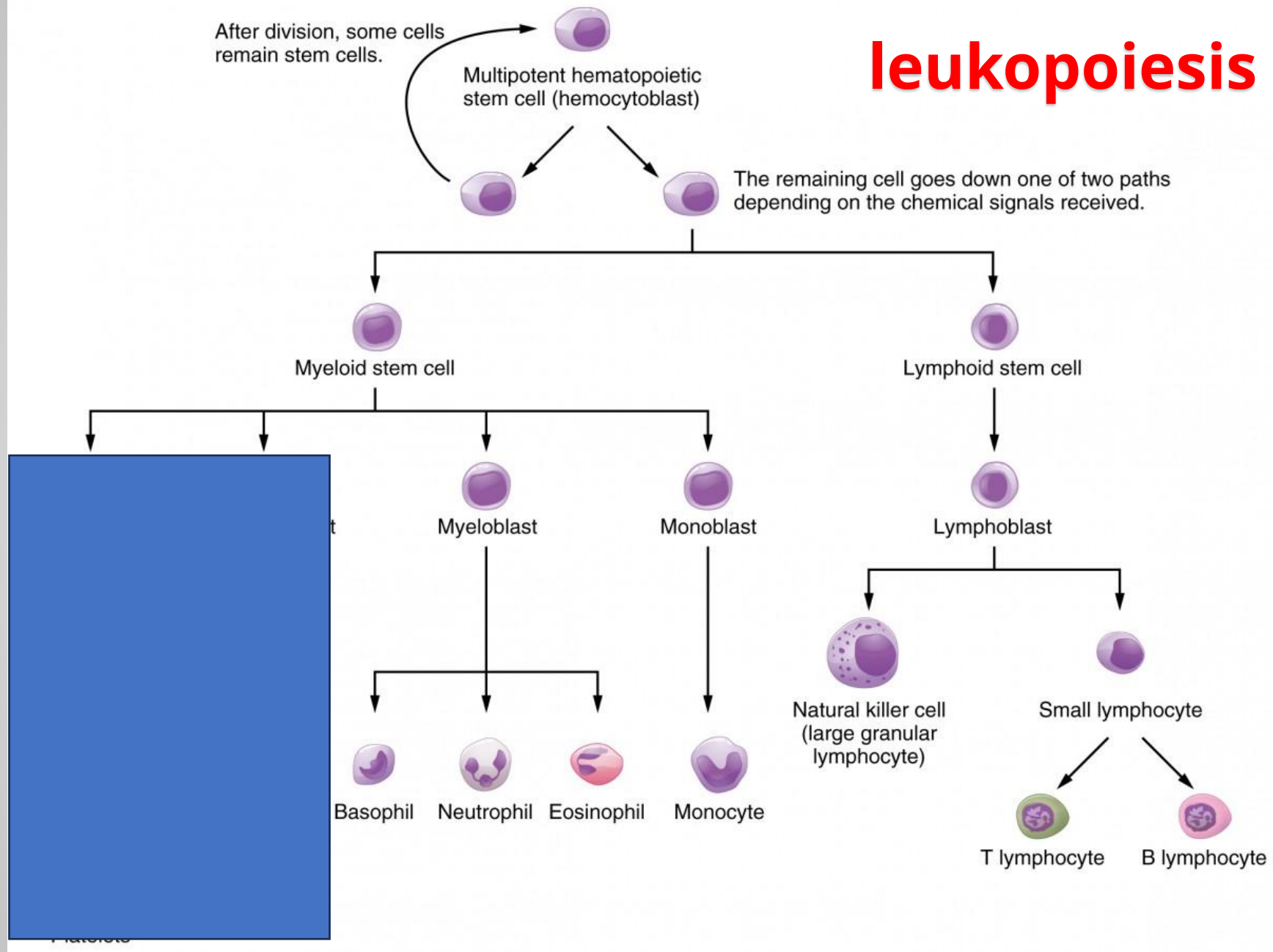
LEUKOCYTE TYPE	PERCENTAGE OF TOTAL WBC (%)	ABSOLUTE COUNT (cells/ μ L)
● Neutrophils	50 – 70	2,000 – 7,000
● Lymphocytes	20 – 40	1,000 – 4,000
● Monocytes	2 – 8	200 – 800
● Eosinophils	1 – 4	50 – 400
● Basophils	0 – 1	0 – 100



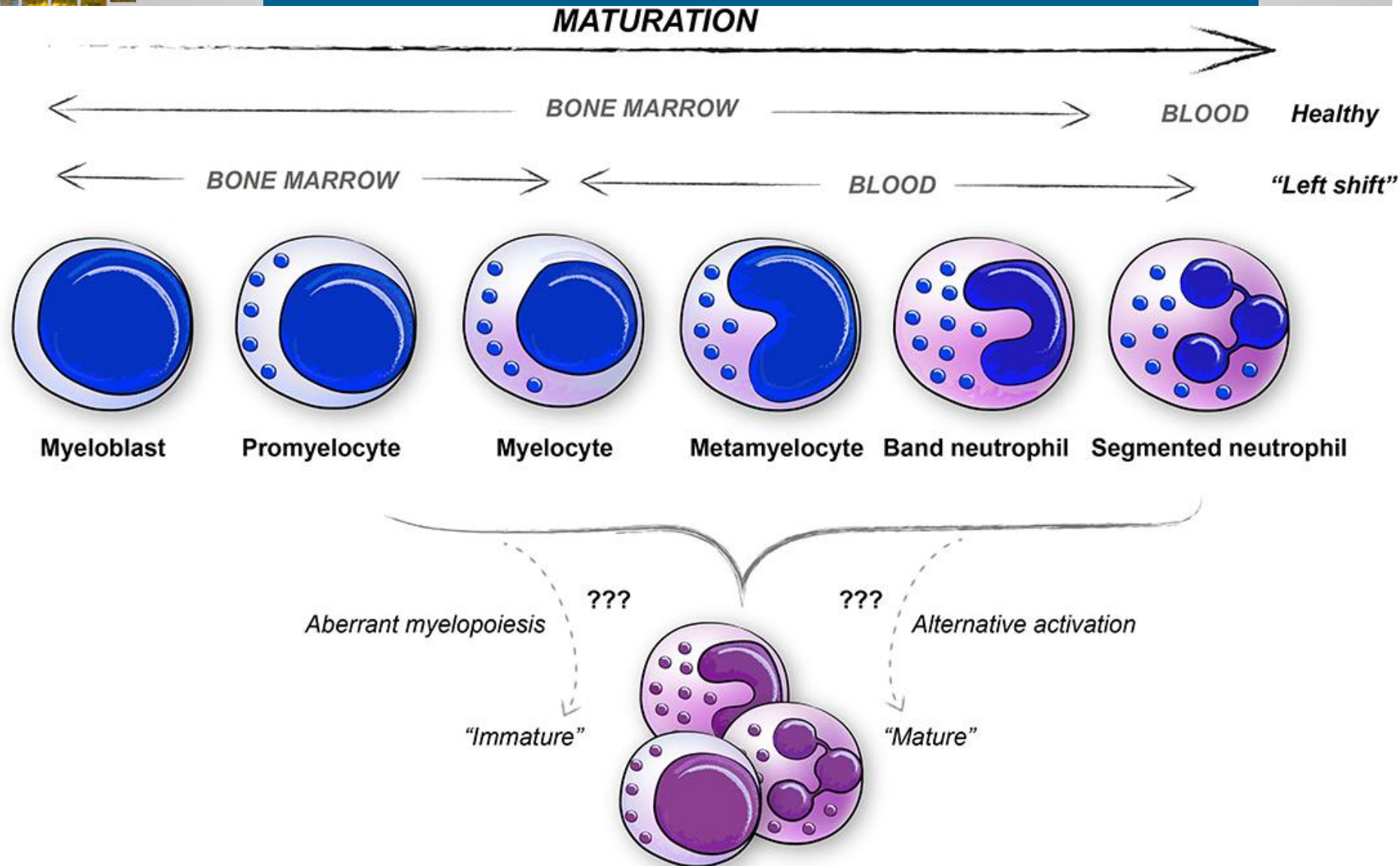
NOTE:

Leukocyte counts may vary with age, sex, physiological conditions, and laboratory methods.

leukopoiesis



Neutrophil Maturation



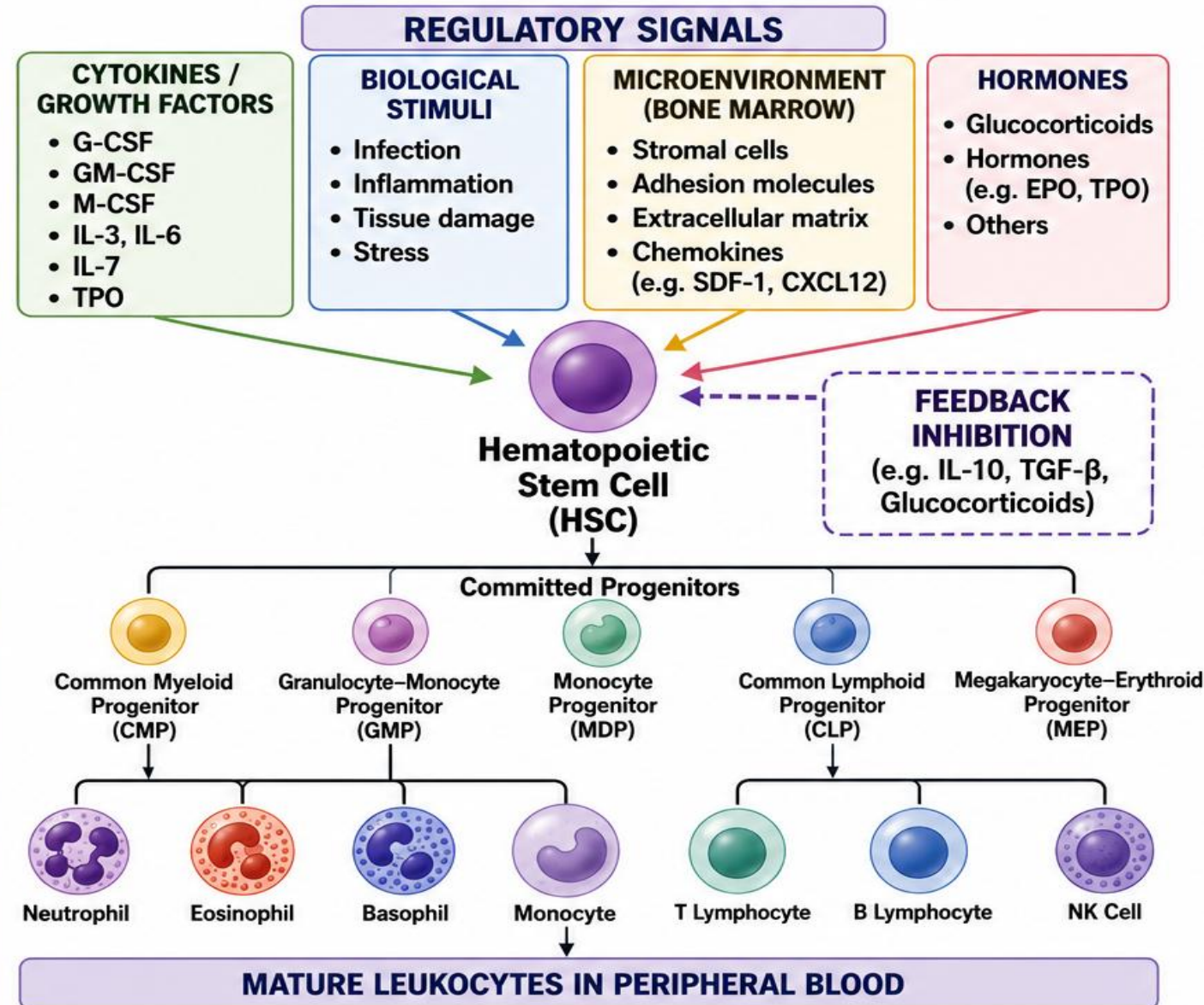
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REGULATION OF LEUKOPOIESIS

Leukopoiesis is the process of white blood cell production in the bone marrow, tightly regulated by cytokines, growth factors, transcription factors, and feedback mechanisms to maintain homeostasis.



Factor	Abbreviation	Molecular weight (kDa)	Target cells	Actions
Granulocyte colony-stimulating factor	G-CSF	18	Granulocyte progenitors, mature neutrophils	Increase neutrophil counts
Granulocyte-macrophage colony-stimulating factor	GM-CSF	14–35	Granulocyte, macrophage progenitors, eosinophil progenitors	Increase neutrophil, eosinophil, and monocyte counts
Erythropoietin	EPO	34–39	Erythroid progenitors	Increase red blood cell counts
Stem cell factor	SCF		Granulocyte-erythroid progenitors, lymphoid progenitors, natural killer cells	Increase pluripotent stem cells and progenitor cells for all other cell types
Thrombopoietin	TPO	35	Stem cells, megakaryocytes, erythroid progenitors	Increase platelet counts
Interleukin-11	IL-11	23	Early hematopoietic progenitors, megakaryocytes	Increase platelet counts

Table 1 ■ Hematopoietic growth factors and their activities.

Table 4. Selected Conditions Associated with Elevations in Certain White Blood Cell Types

<i>White blood cell line</i>	<i>Conditions that typically cause elevations</i>
Basophils	Allergic conditions, leukemias
Eosinophils	Allergic conditions, dermatologic conditions, eosinophilic esophagitis, idiopathic hypereosinophilic syndrome, malignancies, medication reactions, parasitic infections
Lymphocytes	Acute or chronic leukemia, hypersensitivity reaction, infections (viral, pertussis)
Monocytes	Autoimmune disease, infections (Epstein-Barr virus, fungal, protozoan, rickettsial, tuberculosis), splenectomy
Neutrophils	Bone marrow stimulation, chronic inflammation, congenital, infection, medication induced, reactive, splenectomy



Leukopenia

**Bone marrow
failure**

**Inhibition
production
by infection,
drugs, toxin,
etc**

malignancy

**Autoimmune
disease**

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Leukocytosis

inflammation

infection

malignancy

allergy

Injury/
trauma

stress

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